AN IRRATIONALITY MEASURE FOR CATALAN'S CONSTANT

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ABSTRACT. Let $G = \sum_{j\geq 0}(-1)^j(2j+1)^{-2}$ be Catalan's constant. Building on OpenAI's recent proof that G is irrational, we prove that $|G - a/b| > b^{-532}$ for every integer a and every sufficiently large integer b; thus the irrationality exponent of G is at most 532. OpenAI's proof compares, for a determinant Δ whose entries are linear forms in 1 and G, a lower bound coming from the denominators of Δ with an upper bound coming from an integral representation. The determinant depends on five integer parameters, a *design*. We show that the denominator of the determinant is smaller at a prime p by the dimension of the kernel of the reduction modulo p of the matrix of rational parts, and we bound this dimension from below. For an arbitrary design we prove closed formulas on four windows of primes, and a general counting inequality: a space of row combinations on which a few linear conditions force the column responses to vanish, after a difference of step p in the spirit of Lucas's theorem, contributes its dimension minus the rank of the conditions. Instances of this inequality give further families of kernel vectors, and for finitely many windows the inequality is applied to a fixed rational subspace whose required ranks are certified by stored integer kernel vectors. With the design $(58, 9, 3, 5, 3)$, a Gauss quadrature formula and an identity of Andréief type with fixed points, which bound the determinant on an exponentially small neighbourhood of G, and a criterion with a free lower end of the window of primes, this gives the measure. All steps are formalized in the Lean proof assistant, with only the standard axioms.

CONTENTS

1. Introduction 3
1.1. Background and related results 3
1.2. What a measure requires 5
1.3. New ideas 6
1.4. The constants 7
1.5. Organization and notation 7
2. Designs, rank deficiencies and the exponent 8

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2.1. Designs 8
2.2. Rank deficiencies 8
2.3. Valuations, denominators and the exponent 10
2.4. A reading of the exponent 11
3. OpenAI's construction for a design 12
3.1. The determinant 12
3.2. The degree 13
3.3. Rank deficiency and valuation 13
3.4. The digit formula 13
3.5. Integrality and the valuation at p 14
3.6. The denominators 14
3.7. The prime 2 15
3.8. The common denominator 16
3.9. Nonvanishing 16
4. Rank deficiencies modulo p 17
4.1. Setting 18
4.2. Antisymmetric combinations of rows 18
4.3. A reflection of Chebyshev polynomials 19
4.4. A block of columns: the digits 1 to 5 20
4.5. The second window 22
4.6. The counting inequality 23
4.7. Further families 24
4.8. Stored data for finitely many windows 25
4.9. The function κ for the design D 25
5. The real place 27
5.1. A quadrature formula for the coefficient of the constant 27
5.2. Column expansion and an identity of Andréief type 29
5.3. The energy estimate with fixed points 30
5.4. The trial sequences 32
5.5. Verification of the bounds 32
6. Proof of Theorem 1.1 33
6.1. The criterion 33
6.2. The hypotheses for the design D 34
7. Remarks on the method 35
8. Formalization 35
8.1. Module layout 36
8.2. Certificates 38
Disclosure of AI use 38
Acknowledgements 39
References 39

1. Introduction

Catalan's constant

$$G = \sum_{j=0}^{\infty} \frac{(-1)^j}{(2j+1)^2} = L(2, \chi_{-4}) = 0.91596\,55941\,77219\ldots$$

is the value at $s = 2$ of the Dirichlet L-function of the nontrivial character modulo 4. Its arithmetic has long been studied by analogy with $\zeta(2)$ and $\zeta(3)$, but until recently it was not known to be irrational. In October 2026 OpenAI [26] proved that G is irrational and verified the proof in Lean. (An earlier claimed proof [34] was shown to contain a gap [37].)

The *irrationality exponent* $\mu(\xi)$ of a real number ξ is the infimum of the real numbers μ such that $|\xi - a/b| > b^{-\mu}$ for all integers a and all sufficiently large integers b. Every irrational number has $\mu(\xi) \geq 2$, and almost every real number has $\mu(\xi) = 2$. An *irrationality measure* is an explicit upper bound for $\mu(\xi)$.

Theorem 1.1. *For every integer a and every sufficiently large integer b,*

$$\left| G - \frac{a}{b} \right| > b^{-532}.$$

In particular $\mu(G) \leq 532$.

The proof uses the prime number theorem. The theorem and every result in this paper have been verified in the Lean 4 proof assistant on top of OpenAI's formalization, with no unproved steps and only the standard axioms (Section 8).

1.1. Background and related results.

Irrationality proofs for the values of L-functions at integers follow, almost without exception, the pattern of Apéry's proof for $\zeta(3)$ [1]. One constructs integer linear forms $q_n\xi - p_n$ that tend to zero, with $q_n \neq 0$, faster than the growth of their coefficients would allow for a rational ξ. If $|q_n\xi - p_n| = e^{-\alpha n + o(n)}$ and $|q_n| = e^{\beta n + o(n)}$, then ξ is irrational as soon as $\alpha > 0$, and $\mu(\xi) \leq 1 + \beta/\alpha$. Beukers [2] gave the linear forms for $\zeta(2)$ and $\zeta(3)$ as integrals of Legendre-type polynomials. Hata [15] refined such integrals and introduced the systematic use of the primes that divide all the coefficients of the forms, an idea going back to Chudnovsky. Rhin and Viola [29, 30] found that families of integrals for $\zeta(2)$ and $\zeta(3)$ are permuted by a finite group, and used the invariance of the integrals to divide all their coefficients by a large common factor. This *arithmetic method* gave the sharpest measures for $\zeta(2)$ and $\zeta(3)$, and Marcovecchio [25] applied it to $\log 2$. For π, Zeilberger and Zudilin [39] obtained $\mu(\pi) \leq 7.1032\ldots$ from Salikhov's integrals [33] with an additional arithmetic saving.

Since September 2026 a series of new results in this area has appeared, several of them obtained with AI systems. OpenAI proved $\mu(\pi) = 2$, with a Lean formalization [27], and Liu, Jiang and Zhang proved $\mu(\log r) = 2$ for every rational $r > 0$, $r \neq 1$ [24]. Giraudin lowered the bound for $\mu(\zeta(2))$ to

5.0193784 with two hypergeometric constructions of Zudilin [42] at new parameters [13], and the bound for $\mu(\zeta(3))$ to 5.5138800 by applying the Rhin–Viola method at a new integral direction [14]. Leibtag, Barkan, Kalisch, Weinbaum, Shalyt and Kaminer [22] showed that the classical constructions are one-dimensional trajectories in conservative matrix fields, and by searching over other trajectories obtained $\mu(\zeta(2)) < 4.62675$ and $\mu(\zeta(3)) < 5.4197$. Prellberg [28] proved that $\mathrm{Li}_2(-1/4)$, $\mathrm{Li}_2(1/5)$ and $\mathrm{Li}_2(-1/3)$ are irrational with the integrals of Rhin–Viola and Viola–Zudilin. Fauzan announced a proof that $\zeta(5)$ is irrational, using Hankel determinants [11], and Lai combined this method with simultaneous approximations to show that 1, $\zeta(3)$ and $\zeta(4)$ are linearly independent over $\mathbb{Q}$ [20]. Several of these arguments, like the present one, use determinants of large size rather than single sequences of linear forms; Zudilin's determinantal criterion [43] is an earlier instance of this idea.

For Catalan's constant the analogous linear forms exist but do not decay fast enough. Rivoal and Zudilin [31] proved that at least one of the numbers $\beta(2), \beta(4), \ldots, \beta(2m)$ is irrational for suitable m, where β is the Dirichlet beta function; $G = \beta(2)$ is not singled out. Zudilin [41] found an Apéry-like recurrence for G whose solutions give linear forms that converge to zero, but too slowly to prove irrationality, and [40, 18, 19, 44] describe related constructions and their limits. Eskandari, Murty and Nemoto [10] interpret linear forms in Catalan's constant through mixed motives. Leibtag et al. [22] found further Apéry-like approximations to G in their matrix fields; numerically, the best of them still has a negative approximation exponent, about -0.28.

OpenAI's argument [26] has a different shape. Instead of a sequence of linear forms it uses a determinant Δ of size $48N$ whose entries are rational linear combinations of 1 and G. Determinants of linear forms were used for irrationality in Zudilin's determinantal approach [43]; here the entries are double integrals against the Cauchy kernel $1/(1-ts)$. If G were rational, the denominators of the entries would give a lower bound $|\Delta| \geq e^{-(\mathsf{A}+o(1))n^2}$ with $n = 48N$. An integral representation, of the kind that appears in random matrix theory through Andréief's identity [12], gives the upper bound $|\Delta| \leq e^{-(\mathsf{U}+o(1))n^2}$, and an energy estimate shows $\mathsf{U} > \mathsf{A}$. The margin is small: $\mathsf{U} - \mathsf{A} \approx 1.3 \cdot 10^{-4}$. The nonvanishing of Δ along the primes $N = p$ is proved separately by reduction modulo p.

The first irrationality measure for G was obtained by Jha [16], who proved $\mu(G) \leq 10^8$. Carreira [7] treated the five integer parameters of the construction as free, stated conditions under which OpenAI's argument goes through for general parameters, and obtained $\mu(G) < 2294$ in the revised version of his note; he also formulated the best real-place constant of the method as an equilibrium problem for two measures, and in [6] he records attempts to carry the method to $\beta(4)$. His general theorem, instantiated at the design $(36, 6, 2, 3, 2)$ by a search and certification carried out with Claude Fable 5.1,

gives $\mu(G) < 2401.74$ [3]. Xu [38] reported $\mu(G) \le 727.03$ for the design $(20, 3, 1, 2, 1)$, with an improved trial potential, found by an AI system, in the energy estimate at the real place; his certificate is numerical, in ball arithmetic, and is not formalized. The comparison with Carreira's argument is instructive. The determinants are of the same kind. Carreira makes the determinant small near G by bounding all its minors, as Jha does; here a quadrature formula for the coefficient of the constant is used instead (Section 5). And the present paper uses, for the primes p in a union of windows spread over $(16N/7,\ 53N/2]$, the rank of the reduction of the matrix modulo p; this lowers the denominator of the determinant by more than 2.7 times the margin that remains between the real and the finite places without it (Section 1.3). Eskandari [9] studies rational approximations to G from another point of view. For $L(2, \chi_{-3})$, Calegari, Dimitrov and Tang proved linear independence of 1, $\zeta(2)$ and $L(2, \chi_{-3})$ [4], and their holonomy bounds give $\mu(L(2, \chi_{-3})) \le 24781$ [5, Thm. 5.1].

The numbers $\zeta(2)$, G and $L(2, \chi_{-3})$ are differences of values of the trigamma function ψ' at rational points. For $0 < s < t \le 1$ put

$$\Psi(s,t) = \psi'(s) - \psi'(t) = \sum_{n=0}^{\infty} \Big(\frac{1}{(n+s)^2} - \frac{1}{(n+t)^2} \Big).$$

Then

$$\zeta(2) = \tfrac{1}{2}\Psi(\tfrac{1}{2}, 1), \qquad G = \tfrac{1}{16}\Psi(\tfrac{1}{4}, \tfrac{3}{4}), \qquad L(2, \chi_{-3}) = \tfrac{1}{9}\Psi(\tfrac{1}{3}, \tfrac{2}{3}),$$
$$G + \tfrac{5\pi^2}{48} = \tfrac{1}{8}\Psi(\tfrac{1}{4}, 1), \qquad G - \tfrac{5\pi^2}{48} = -\tfrac{1}{8}\Psi(\tfrac{3}{4}, 1).$$

Since the irrationality exponent does not change under $\xi \mapsto r\xi + r'$ with $r, r' \in \mathbb{Q}$, $r \ne 0$, a measure for one of these numbers is a measure for the corresponding value of Ψ. The companion paper [23] proves measures for $\Psi(\frac{a}{q}, 1)$ with $q \in \{3, 4, 5, 7\}$, $0 < a < q$, $\gcd(a, q) = 1$, and for $\Psi(\frac{1}{3}, \frac{2}{3})$, from Löwner matrices of digamma differences, with a method related to the present one.

1.2. **What a measure requires.** Write $\Delta_N(X)$ for the determinant regarded as a polynomial in the constant, so that $\Delta_N(G)$ is OpenAI's determinant. Three properties of the integer polynomials Q_N obtained from Δ_N by clearing denominators turn the comparison of OpenAI into a measure.

(i) *Smallness on a neighbourhood of G.* The upper bound must hold not only at $X = G$, but for all X with $|X - G| \le e^{-\rho N}$, and the exponent ρ enters the final bound linearly.

(ii) *Uniform nonvanishing.* $Q_p(a/b) \ne 0$ for primes $p \nmid b$ (here $p^2 \nmid b$), in place of $Q_p(G) \ne 0$.

(iii) *Arithmetic.* The integer $b^{\deg Q_N} Q_N(a/b)$ is nonzero, hence at least 1 in absolute value; the smaller the denominator of Δ_N and the smaller its degree, the stronger the conclusion. The prime p must be chosen not dividing b, and a naive choice costs a factor.

With the degree d of Q_N in X, the decay rate η of Q_N (normalised by N^2) and ρ, the conclusion is $\mu(G) \leq d\rho/\eta$ (Proposition 2.8). Everything below serves to make d and ρ small and η large for a design.

1.3. **New ideas.**

1. *All designs at once.* The determinant has five integer parameters $(n_0, b_0, q_0, g_0, h_0)$, a *design*, and every argument below is carried out for an arbitrary design, with the parameters as symbols: the degree $n - g$, the denominators at all primes, the rank deficiencies, the nonvanishing and the real-place budgets. A design is then data. We use $(58, 9, 3, 5, 3)$, so that $n = 58N$, $b = 9N$, $q = 3N$, $g = 5N$, $h = 3N$ and the degree is $53N$.

2. *Rank of the reduction modulo p, in closed form.* Let $\nu_N(p)$ be the dimension of the kernel (on the left) of the reduction modulo p of p^2 times the matrix of rational parts. We prove four lower bounds for $\nu_N(p)$ (Theorem 2.1), each a piecewise linear function of p with coefficients that are linear in the five parameters, each on its own explicit window of primes: antisymmetric combinations of rows centred at g, $g + p$ and $g + 2p$ (Theorem 4.1); a reflection of Chebyshev polynomials modulo p (Theorem 4.3); and two block arguments on columns, using Frobenius relations $(1 - X)^p = 1 - X^p$ for the coefficients of the row polynomials (Theorems 4.4 and 4.6). They are proved from one digit formula for the entries modulo p (Lemma 3.3). For $(58, 9, 3, 5, 3)$ they cover the primes $12N < p \leq 26N$.

3. *A counting inequality for the nullity.* The four proofs have a common shape: a space $\mathcal{V}$ of row combinations, a few linear conditions on $\mathcal{V}$, and an argument that the conditions force all column responses to vanish. We prove this once (Theorem 2.2): if the conditions Φ on $\mathcal{V}$ force the responses on the raw columns to vanish after the $\lfloor q/p \rfloor$-fold difference of step p, up to a polynomial of degree less than q mod p, then $\nu_N(p) \geq \dim \mathcal{V} - \operatorname{rank} \Phi$. The step of size p comes from the congruence $(1 - T)^q \equiv (1 - T)^{q \bmod p}(1 - T^p)^{\lfloor q/p \rfloor} \pmod p$ of Lucas's theorem, so the inequality also applies when the filter is longer than the prime. Further families of kernel vectors are instances of it, for every design: Frobenius products, in which the row polynomial is a fixed polynomial times a polynomial in X^p, the conditions being Taylor coefficients at $X = \pm 1$, coefficients at 0 and at ∞ and finitely many responses; and families in which the responses are constant along the digit $\lfloor j/p \rfloor$ of the column, which then cost only $p - q$ conditions (Section 4.7).

4. *A fixed rational subspace, with stored kernel vectors.* In the *profile form* of the inequality the space is $\Pi \otimes V$, where Π is a space of m polynomial coordinates and V is a fixed subspace of $\mathbb{Q}^s$ of dimension d, and the conditions come in groups of w_j jets composed with fixed linear maps B_j on V. Then $\nu_N(p) \geq m\,d - \sum_j w_j \operatorname{rank}(B_j|_V)$, where m and the w_j are affine in N and p on a window, while V and the B_j do not depend on N or p (Theorem 2.3). Upper bounds r_j for the ranks are certified by $d - r_j$ integer vectors in the coordinates of V that B_j kills, and by one nonzero minor showing that they stay independent modulo p; the identification of the conditions with the jets

is checked on each window by a verified decision procedure. For $(58, 9, 3, 5, 3)$ this is done for 34 windows between $3N$ and $16.2N$.

5. *One bound per prime.* At a prime p the valuation of $\Delta_N(z)$ is at least $\nu_N(p)-2n$ for every z that is p-integral (Proposition 2.5); the denominator of the determinant is therefore divided by $p^{\nu_N(p)}$, and on each window of primes one bound is used. No separate common factor of the coefficients is needed: integer-valuedness of Δ_N at the integers already makes the coefficients of Q_N divisible by what is gained (Proposition 2.6). For $(58, 9, 3, 5, 3)$ the windows give a piecewise linear profile κ with 102 pieces on $(16/7, 53/2]$, and the gain is $\int \kappa(x)\,dx = 22.05972\ldots$ in units of N^2 (Theorem 2.4).

6. *Perturbation by quadrature with an odd number of nodes.* The smallness on a neighbourhood of G is obtained from a Gauss-type quadrature formula for the coefficient of the constant, with $d_0N = 53N$ nodes (an odd number for odd N) kept at a distance from the origin, and an identity of Andréief type with fixed points (Section 5).

7. *Trial potentials with complex poles.* The energy estimate at the real place uses trial sequences whose generating functions are logarithmic potentials of point charges at complex points close to the edges of the support of the extremal measure, and a near-field estimate at the first Chebyshev coefficient (Section 5).

8. *A criterion with a free lower end of the window.* In the choice of the prime, the window of candidates is $(V, U]$ with $U = \kappa \log b$ and $V = (\kappa - \omega)\log b$, and the natural choice $\omega = 1$ requires $\kappa > 2$. We prove the criterion for every $\omega \in (0, 1]$ with the hypothesis $\kappa > 2\omega$ (Proposition 2.8); the design above has $\kappa = 1.764261$ and uses $\omega = 3/4$.

1.4. **The constants.** For the design $(58, 9, 3, 5, 3)$, in the unit N with $n = 58N$: the degree of Q_N is $d = 53N$; the perturbation exponent is $\rho = 301.54258$; the decay rate of Q_N is $\eta = 30.041$, a rational number below the sum of the real place minus the denominators without the gain, $7.987764\ldots$, and the gain from the rank deficiencies, $\int \kappa = 22.05972\ldots$; the proof uses the lower bound $\int \kappa \geq 22.05325$. In the criterion, $\omega = 3/4$, $\delta = 2.9 \cdot 10^{-7}$ and $\kappa = 532/\rho = 1.764261\ldots$ Decimals are truncated; the constants of the formal proof are the exact rational numbers. With $\eta = 30.041$ the last condition of the criterion reads $B > 531.99924\ldots$, and the others hold with room to spare (Section 6); the integer 532 is the exponent of Theorem 1.1, for which all the conditions are verified in Lean with rational constants.

1.5. **Organization and notation.** Section 2 sets up designs and states the rank deficiencies, the valuations, the denominators and the criterion, which are the new ideas in precise form. Section 3 recalls OpenAI's construction for a design and proves the degree, the denominators and the nonvanishing. Section 4 proves the rank deficiencies. Section 5 contains the estimates at the real place, Section 6 proves Theorem 1.1, and Section 7 makes remarks on the method. Section 8 describes the formalization; the disclosure of AI use follows it.

We use the notation of OpenAI's paper [26] for the construction and refer to it for details; citations of the form [26, Prop. 2.1] point to its numbered statements. We write log for the natural logarithm, v_p for the p-adic valuation, $\theta(x) = \sum_{p \le x} \log p$ for Chebyshev's function, T_k and U_k for the Chebyshev polynomials of the first and second kind (with $U_{-1} = 0$ and $T_{-k} = T_k$, $U_{-k-1} = -U_{k-1}$), $[X^k]F$ or $F[k]$ for the coefficient of X^k in a polynomial F, $y_+ = \max(y, 0)$, and $\operatorname{rev}_E F(X) = X^E F(1/X)$ for the reversal of a polynomial of degree at most E.

2. Designs, rank deficiencies and the exponent

2.1. **Designs.** A *design* is a quintuple $(n_0, b_0, q_0, g_0, h_0)$ of integers with $n_0, b_0, q_0, h_0 \ge 1$, $g_0 \ge 0$ and $2g_0 < n_0$. For $N \ge 1$ we write

$$n = n_0 N, \quad b = b_0 N, \quad q = q_0 N, \quad g = g_0 N, \quad h = h_0 N,$$
$$L = n + b + q, \quad C = L + g, \quad H = C + h, \quad A = b + q + 2g,$$

and $d_0 = n_0 - g_0$. OpenAI's choice is the design $(48, 7, 4, 4, 2)$. We work throughout with

$$D = (58, 9, 3, 5, 3): \quad n = 58N, \quad b = 9N, \quad q = 3N, \quad g = 5N, \quad h = 3N,$$
$$L = 70N, \quad C = 75N, \quad H = 78N, \quad A = 22N,$$

and $d_0 = 53$. The construction of OpenAI's determinant for a design is recalled in Section 3; it yields an $n \times n$ matrix $\mathcal{A}_N = \mathcal{A}_N^0 + G\mathcal{L}_N$ with rational matrices $\mathcal{A}_N^0$ and $\mathcal{L}_N$, and the polynomial $\Delta_N(X) = \det(\mathcal{A}_N^0 + X\mathcal{L}_N)$ of degree at most $n - g$ (Lemma 3.1). The search that led to D among designs, scored by the exponent of Proposition 2.8, is not part of the proof.

2.2. **Rank deficiencies.** For an odd prime p with $H < p^2$ the entries of $p^2\mathcal{A}_N^0$ are p-integral (Lemma 3.3); let $\mathcal{A}_{N,p}^0$ be its reduction modulo p, and let $\nu_N(p)$ be the dimension of its left kernel, that is n minus its rank. Write $\Lambda = g + q + h$ and $e = (p+1)/2 - q - h$.

Theorem 2.1. *Let D be a design, $N \ge 1$, and p an odd prime with $H < p^2$. Then $\nu_N(p)$ is at least each of the following quantities, whenever the stated conditions hold; negative values are replaced by 0.*

(a) (Theorem 4.1) If $2g < p$, $H \le 4p$ and $D_T = \min(g,\, n - 1 - g - 2p,\, C - 3p) \ge 1$:

$$D_T - (n + q + g + h - 3p)_+.$$

(b) (Theorem 4.3) If $2h \le p + 1$, $L \le 3p \le H$, $n + q + g + h \le 3p$ and $2p \le n + q$:

$$\min(n,\, 2p + 2g + 1) - g - 2p - 1.$$

(c) (Theorem 4.4) If $e \ge 0$, $n \le 4p + 2g + 1$, $L \le 6p$ and $b + q \le p + \mu_0$, with $\mu_0 = \max(q, H + q - 6p)$:

$$\min\Big(\frac{p-1}{2} - g,\; n + b + q + h - \frac{11p+1}{2}\Big).$$

(d) (Theorem 4.6) If $e \geq 0$ and $n \leq 4p + 2g + 1$: $m_1 - m_0 - e$, with

$$m_0 = \max(b + q,\ L + \Lambda - 4p), \qquad m_1 = \min(p,\ L - 3p,\ b + q + 2g + 1).$$

These are the statements of the Lean theorems `t_rank`, `u_rank`, `u6_rank_closed` and `s_rank` (the third in the closed form of (c), the fourth in the form $m_1 - m_0 - e$), proved for every admissible design. For D they apply for $12N < p \leq 26N$, where they give most of the gain (Section 4.9).

Below $12N$ the closed formulas give nothing, and further kernel vectors come from a single counting inequality, of which the proofs of Theorem 2.1 are special cases. For $c \in \mathbb{F}_p^n$ and a raw column $j \in [b, L)$ let $R_c(j)$ be the reduction modulo p of $p^2 \sum_r c_r R(r, j)$, with R as in (3.4); these are the responses of Lemma 3.3.

Theorem 2.2. *Let D be a design, $N \geq 1$, p an odd prime with $H < p^2$, and write $q = ap + r$ with $0 \leq r < p$. Let $\mathcal{V}$ be a finite-dimensional vector space over $\mathbb{F}_p$, $\iota\colon \mathcal{V} \to \mathbb{F}_p^n$ an injective linear map and Φ a linear map on $\mathcal{V}$. Suppose that for every $x \in \ker \Phi$ there is a polynomial $G_x \in \mathbb{F}_p[X]$ of degree less than r with*

$$\sum_{v=0}^{a} (-1)^{a-v} \binom{a}{v} R_{\iota(x)}(j + vp) = G_x(j) \qquad (b \leq j < L - ap).$$

Then $\nu_N(p) \geq \dim \mathcal{V} - \operatorname{rank} \Phi$.

For $q < p$ the hypothesis says that the conditions Φ force all raw responses to vanish. The case of interest for the counting is the *profile form*, in which the space is a tensor product of a space of polynomial coordinates with a fixed rational subspace, and the conditions are jets composed with fixed linear maps.

Theorem 2.3. *Let $v_1, \dots, v_d \in \mathbb{Z}^s$ span a subspace $V \subset \mathbb{Q}^s$ of dimension d, and for j in a finite set J let B_j be an integer matrix with s columns and $r_j \leq d$ an integer such that there are $d - r_j$ vectors $k \in \mathbb{Z}^d$, linearly independent, with $B_j(\sum_a k_a v_a) = 0$. Let $\mathcal{E} \neq 0$ be the product of a nonzero $d \times d$ minor of $(v_1, \dots, v_d)$ and, for each j, of a nonzero maximal minor of these kernel vectors. Let $p \nmid \mathcal{E}$ be an odd prime with $H < p^2$, write $q = ap + r$ as above, let $V_p = \mathbb{F}_p^d$ be the space of coordinates of V modulo p, and let m and w_j ($j \in J$) be integers ≥ 0. Suppose that there are linear maps $\varphi_{j,k}\colon V_p^m \to V_p$ ($k < w_j$) and $\iota\colon V_p^m \to \mathbb{F}_p^n$ such that, on the joint kernel of the maps $B_j \circ \varphi_{j,k}$, the map ι is injective and satisfies the hypothesis of Theorem 2.2. Then*

$$\nu_N(p) \geq m\, d - \sum_{j \in J} w_j\, r_j.$$

In the applications m and the w_j are affine functions of N and p on a window of primes, while V and the B_j are fixed; so the bound is linear on the window, and the data to be stored are the vectors v_a, the matrices B_j and the kernel vectors. For the design D these bounds, the closed formulas and the further families of Section 4 combine into the following.

Theorem 2.4. *For the design D there are 102 intervals $(a_i, b_i]$, with $16/7 = a_1 < b_1 \le a_2 < \cdots < b_{102} = 53/2$, and a function κ, linear on each interval and 0 outside their union, such that for all large N and every prime p with $a_i N < p \le b_i N$ for some i*

$$\nu_N(p) \ge N\kappa(p/N) - 2.$$

On each interval the bound is given by one of the families of Section 4. The area of κ is a rational number,

$$\int_0^\infty \kappa(x)\,dx = \frac{362355158746019386015965 98}{16426096632711159925781 25} = 22.05972\ldots \;\ge\; 22.05325.$$

The contributions by ranges of $x = p/N$ are as follows; the third column counts the intervals on which the bound is that of Theorem 2.3 with stored data.

range of x	intervals	with stored data	area
(16/7, 3]	11	0	0.06259
(3, 6]	46	23	0.25136
(6, 12]	27	9	0.71965
(12, 53/2]	18	2	21.02610
total	102	34	22.05972

The intervals in $(12, 53/2]$ are listed in Section 4.9, and the full list is part of the formalization (Section 8). On each interval one bound is used. The families come from different arguments, and except for one pair (Section 4.7) the kernels they produce are not shown to be independent, so at a prime the bounds are not added.

2.3. **Valuations, denominators and the exponent.** The rank deficiencies enter through the following statement, which is the finite-place input.

Proposition 2.5. *Let p be an odd prime with $H < p^2$ and let $z \in \mathbb{Q}$ have denominator prime to p. Then $p^2(\mathcal{A}_N^0 + z\mathcal{L}_N)$ has p-integral entries, its reduction modulo p equals $\mathcal{A}_{N,p}^0$, and*

$$v_p(\Delta_N(z)) \ge \nu_N(p) - 2n.$$

At the other primes there are the valuation bounds of OpenAI's method (Section 3), and at 2 a bound from the expansion into minors. They combine to the denominators.

Proposition 2.6. *For the design D there are positive integers D_N such that $D_N\Delta_N(i) \in \mathbb{Z}$ for every integer i, such that $q^{2n} \mid D_N$ for every prime $N < q \le 2N$, and such that for every $\varepsilon > 0$ and all large N*

$$\log D_N \le (\mathsf{A} + \varepsilon)N^2,$$

$$\mathsf{A} = \frac{2673}{8}\log 2 + 116 + \frac{24269}{4} - \int_0^\infty \kappa(x)\,dx = 6392.7880\ldots$$

Here $2673/8\log 2$ is the contribution of the prime 2, and $116 + 24269/4$ is the area of the piecewise linear function that bounds the poles at the odd primes (Section 3); the last term is the gain $\int \kappa$ of Theorem 2.4.

At the real place we prove the following in Section 5. Put

$$\rho = n_0 c_2 + w, \qquad \rho = 301.54258 \ \text{ for } D, \tag{2.1}$$

where $c_2 = 4.28886$ is the cost of a fixed column in the energy estimate and $w = 52.7887$ is the logarithmic size of the weights of the quadrature formula.

Theorem 2.7. *Let $U = 2.255858836$. For every $\varepsilon > 0$, all large N and all real x with $|x - G| \le e^{-\rho N}$,*

$$|\Delta_N(x)| \le \exp\bigl(-(n_0^2(U - \tfrac{1}{2}\log 2) - \varepsilon)N^2\bigr), \qquad n_0 = 58.$$

Since $n_0^2(U - \frac{1}{2}\log 2) = 6422.8355\ldots$ and $\mathsf{A} = 6392.7880\ldots$, the polynomial $Q_N = (d_0 N)!\, D_N \Delta_N$ has integer coefficients, degree at most $d_0 N$, and

$$|Q_N(x)| \le e^{-\eta N^2} \qquad (|x - G| \le e^{-\rho N}), \qquad \eta = 30.041,$$

where η is a rational number slightly below $n_0^2(U - \frac{1}{2}\log 2) - \mathsf{A} = 30.04748\ldots$ The two remaining properties of Section 1.2 are the nonvanishing of $Q_p(a/b)$ for $p^2 \nmid b$ (Proposition 3.5) and the divisibility of the i-th coefficient of Q_N by q^{2i} for $N < q \le 2N$ (Lemma 3.4), which come from the denominators and are what is used to choose the prime. The criterion is the following.

Proposition 2.8. *Let $\xi \in \mathbb{R}$, $d \in \mathbb{N}$, and let $\eta, \rho, \kappa, \delta, \omega > 0$ satisfy*

$$\omega \le 1, \quad 2\omega < \kappa, \quad \eta < 2d, \quad 1 < 2\bigl(\omega - \delta(2\kappa - \omega)\bigr), \quad d + 4d\delta\kappa < \eta\kappa.$$

Let $Q_p \in \mathbb{Z}[X]$ (p prime) satisfy, for all large p: $\deg Q_p \le dp$; $Q_p(a/b) \ne 0$ whenever $a \in \mathbb{Z}$, $b \ge 1$, $p^2 \nmid b$; q^{2i} divides the i-th coefficient of Q_p for all primes $p < q \le 2p$ and $i \le dp$; and $|Q_p(x)| \le e^{-\eta p^2}$ for $|x - \xi| \le e^{-\rho p}$. Then

$$|\xi - a/b| > b^{-\rho\kappa}$$

for all integers a and all sufficiently large b.

The conditions say, in the order of the proof of Section 6, that the window of primes contains a prime not dividing b, that the primes of the window above the chosen one are recycled, and that the final inequality is contradictory. For the design D one takes $d = 53$, $\omega = 3/4$, $\delta = 2.9\cdot 10^{-7}$ and $\kappa = B/\rho$ with $B = 532$; the last condition then reads $B > x := d\rho/(\eta - 4d\delta) = 531.99924\ldots$

2.4. **A reading of the exponent.** The criterion shows that, for a fixed design, the bound is $d\rho/\eta$ in the limit, and that its three ingredients come from three different places: the degree $d = n - g$ from the construction, the decay rate η from the real place, the denominators and the rank deficiencies, and the perturbation exponent ρ from the quadrature. The rank deficiencies matter because η is a small difference of two large numbers, 6422.8355 and 6392.7880; the gain 22.06 is nearly three quarters of $\eta = 30.041$.

3. OpenAI's construction for a design

We recall the construction of [26] in the generality in which we use it, and we prove for an arbitrary design the degree bound, the bounds for the denominators at the finite places and the nonvanishing at the primes $N = p$. Carreira [7] checks that the argument of OpenAI goes through for designs satisfying a few conditions; we need the same for the finite places and, in addition, for the entries modulo p^2.

3.1. **The determinant.** Let $D = (n_0, b_0, q_0, g_0, h_0)$ be a design, with the notation of Section 2. For $0 \le r < n$ the row polynomials are

$$P_r(t) = (1-t)^h t^{C-1} T_{|r-g|}(1/t), \qquad \deg P_r < H, \tag{3.1}$$

together with companion polynomials D_r built from $U_{|r-g|-1}$ [26, (3)]. Thus P_r depends only on $|r-g|$. Put $f(t) = \sqrt{1-t^2}$ and $d\mu(t) = |t|\,dt/f(t)$ on $(-1,1)$, and

$$A_r(t) = P_r(t) - \frac{3}{2}\,\mathbf{1}_{\{t>0\}}\frac{f(t)}{t} D_r(t), \qquad \psi_k(s) = s^{b+k}(1-s)^q.$$

The determinant studied by OpenAI is that of the $n \times n$ matrix

$$\mathcal{A}_N(r,k) = \int_{-1}^{1}\int_{0}^{1} A_r(t)\,\frac{\psi_k(s)}{1-ts}\,ds\,d\mu(t) \qquad (0 \le r,k < n). \tag{3.2}$$

The factor $(1-s)^q$ in ψ_k amounts to a finite difference of order q applied to the columns. The entries are rational linear combinations of 1 and G: there are rational matrices $\mathcal{A}_N^0$ and $\mathcal{L}_N$ with

$$\mathcal{A}_N = \mathcal{A}_N^0 + G\,\mathcal{L}_N, \qquad \mathcal{L}_N(r,k) = \sum_{v=0}^{q}(-1)^v\binom{q}{v}\sum_i [t^i]P_r \cdot 4\,c_{(i-b-k-v)/2}, \tag{3.3}$$

where $c_l = 4^{-l}\binom{2l}{l}$ for integers $l \ge 0$ and $c_z = 0$ otherwise [26, Prop. 2.1 and (20)]. The rational part is

$$\begin{aligned} \mathcal{A}_N^0(r,k) &= \sum_{v=0}^{q}(-1)^v\binom{q}{v} R(r, b+k+v), \\ R(r,j) &= \sum_{i<H} P_r[i]\,M(i,j) - \frac{3}{2}\sum_{i<H} D_r[i]\,Z(i,j) \end{aligned} \tag{3.4}$$

with rational numbers $M(i,j)$ (moments of μ against the Cauchy kernel) and $Z(i,j)$, where $Z(i,i) = -\sum_{k\le i} k^{-2}$ and $Z(i,j) = (H_i - H_j)/(i-j)$ for $i \ne j$, $H_i = \sum_{k\le i} 1/k$ [26, Prop. 2.1]. The design enters (3.4) only through the integer coefficients $P_r[i], D_r[i]$ and the range of j, which is $[b, L)$. Put $\Delta_N(X) = \det(\mathcal{A}_N^0 + X\mathcal{L}_N) \in \mathbb{Q}[X]$, so that $\Delta_N(G) = \det \mathcal{A}_N$.

3.2. **The degree.**

Lemma 3.1. $\deg \Delta_N \le n - g$.

Proof. By (3.3) the row r of $\mathcal{L}_N$ depends only on P_r, and hence only on $|r-g|$. So the rows r and $2g-r$ of $\mathcal{L}_N$ agree for $0 \le r < g$, and $2g - r \le n-1$ because $2g < n$. Subtracting row r from row $2g-r$ for each $r < g$ does not change the determinant and makes g rows of the new coefficient of X vanish. A determinant $\det(B + XL')$ in which L' has $n-g$ nonzero rows has degree at most $n-g$ in X. □

3.3. **Rank deficiency and valuation.** We use repeatedly the following elementary fact.

Lemma 3.2. *Let M be an $n \times n$ matrix with entries in $\mathbb{Z}_{(p)}$ whose reduction modulo p has rank at most $n-e$. Then $v_p(\det M) \ge e$.*

Proof. Factor the reduction through $\mathbb{F}_p^{n-e}$ and lift the two factors to integral matrices U, V, so that $M = UV + pE$ with E integral at p. Expanding the determinant multilinearly in the columns, every term that uses more than $n-e$ columns of UV vanishes, and every other term has at least e factors p. □

3.4. **The digit formula.** The reduction of $p^2\mathcal{A}_N^0$ modulo p is described by the digits in base p of the coefficients of the row polynomials. This is the one place where the arithmetic of the moments enters; it is design-free. For an odd prime p and $c \in \mathbb{F}_p^n$ put

$$P_c = \sum_r c_r P_r, \qquad D_c = \sum_r c_r D_r, \qquad W = (1-X^2)^{(p-1)/2},$$

(reading the P_r, D_r as polynomials in X), and let $F[k]$ denote the coefficient of X^k.

Lemma 3.3 (digit formula)**.** *Let p be an odd prime with $H < p^2$, let $0 \le j < H$, and write $j = Jp + \ell$ with $0 \le \ell < p$. Then $p^2 R(r,j)$ is p-integral for every row r, and for every $c \in \mathbb{Z}^n$*

$$p^2 \sum_r c_r R(r,j) \equiv \sum_{u \ge 0} \Big(\alpha_{u,J}\, (XWP_c)[up+\ell] + \beta_{u,J}\, D_c[up+\ell] \Big) \pmod p,$$

where $\alpha_{u,J} = M(u-1,J)$ and $\beta_{u,J} = -\frac{3}{2} Z(u,J)$ are p-integral numbers that do not depend on c, on ℓ or on the design.

Proof. Two facts about the numbers M and Z are needed, for $0 \le i, j < p^2$. Write $j = Jp + \ell$ and let $\varrho \in [0,p)$ be the residue of $j - i - 1$ modulo p, $u = (i + 1 + \varrho - \ell)/p$. First, $p^2 M(i,j)$ is p-integral and $p^2 M(i,j) \equiv W[\varrho]\, M(u-1, J) \pmod p$; this is the digit reduction of the moments in [26, §3], obtained from the p-adic analysis of their recursions, and it does not involve a design. Second, $p^2 Z(i,j)$ is p-integral and is congruent to $Z(\lfloor i/p \rfloor, J)$ if $i \equiv j \pmod p$ and to 0 otherwise. For the second, write

$i' = \lfloor i/p \rfloor$. Since $i < p^2$, $H_i = p^{-1}H_{i'} + S_i$ and $H_i^{(2)} = p^{-2}H_{i'}^{(2)} + S_i'$ with S_i, S_i' p-integral, which gives the case $i = j$. If $i \not\equiv j$, then $p^2 Z(i,j) = p \cdot pZ(i,j) \equiv 0$. If $i \equiv j$, $i \neq j$, then $i - j = (i' - J)p$, and the integers $k \in (j, i]$ prime to p form $i' - J$ complete systems of residues modulo p, so that $\sum_k 1/k \equiv (i'-J)\sum_{r=1}^{p-1} 1/r \equiv 0 \pmod p$; hence $H_i - H_j \equiv (H_{i'} - H_J)/p$ and $p^2 Z(i,j) \equiv Z(i', J)$.

Now use (3.4). By the first fact the part $\sum_i F[i]\, p^2 M(i,j)$ with $F = P_c$ is $\sum_i F[i]\, W[\varrho(i)]\, M(u(i) - 1, J)$. For fixed u the indices with $u(i) = u$ are those for which $\varrho = up + \ell - 1 - i$ lies in $[0, p-1]$, and W has degree $p-1$, so the sum over these i is $(XFW)[up+\ell]$; this gives the α-terms. By the second fact the part with D_c is $\sum_{i \equiv \ell} D_c[i]\, Z(\lfloor i/p \rfloor, J) = \sum_u Z(u,J) D_c[up+\ell]$, which gives the β-terms. The p-integrality of $M(u-1, J)$ and $Z(u, J)$ for $u, J < p$ is part of the first and second facts. □

For the filtered columns we use the following form. Let $0 \le k < n$, $s = b + k$, $J = \lfloor s/p \rfloor$ and $\mu = (s \bmod p) + q$. If $\mu < p$, the $q+1$ raw columns $s, \dots, s+q$ that enter column k have the same digit J and residues $(s \bmod p) + v < p$. Since $\sum_{v=0}^{q} (-1)^v \binom{q}{v} F[t+v] = [X^{t+q}]((X-1)^q F)$ for every polynomial F, Lemma 3.3 gives

$$p^2 \sum_r c_r \mathcal{A}_N^0(r,k) \equiv \sum_{u \ge 0} \Big(\alpha_{u,J}\, Q_c[up + \mu] + \beta_{u,J}\, \widehat{D}_c[up + \mu] \Big), \tag{3.5}$$

where $Q_c = (X-1)^q\, XW P_c$ and $\widehat{D}_c = (X-1)^q D_c$. A combination c whose responses vanish on all raw columns in $[b, L)$ lies in the left kernel of $\mathcal{A}_{N,p}^0$, and a combination whose responses on the columns outside a set vanish and which kills the set by (3.5) does as well.

3.5. **Integrality and the valuation at p.**

Proof of Proposition 2.5. By Lemma 3.3 the entries of $p^2\mathcal{A}_N^0$ are p-integral, since the filter is an integer combination of raw columns with $j < L \le H$. The entries of $\mathcal{L}_N$ are p-integral for odd p, since $P_r[i] \in \mathbb{Z}$ and $c_l \in \mathbb{Z}[1/2]$, so $p^2\mathcal{L}_N \equiv 0 \pmod{p^2}$ and, for z with denominator prime to p, $p^2(\mathcal{A}_N^0 + z\mathcal{L}_N)$ has p-integral entries with the same reduction as $p^2\mathcal{A}_N^0$. By Lemma 3.2 the determinant of the latter has valuation at least $\nu_N(p)$, and $\det\big(p^2(\mathcal{A}_N^0 + z\mathcal{L}_N)\big) = p^{2n}\Delta_N(z)$. □

3.6. **The denominators.**

Let p be an odd prime with $H < p^2$ and $p \le H$. The method of [26, §3], applied to a design, bounds from below the valuation of $\Delta_N(z)$ for every z with denominator prime to p by $-\ell_N(p)$, where $\ell_N(p)$ counts the columns whose entries have a pole at p, once the coefficients of the row polynomials are grouped by their digits in base p and the column filter is taken into account. It is given by two formulas, in which all differences are truncated at 0. For $2p \le H$,

$$\ell_N(p) = 2n - \Big(\min(p,\, L - p,\, A) - \max(b,\, H - 2p) - q \Big),$$

and for $H < 2p \le 2H$, with $R = (L-b) - (\min(A, L-p) - b) - (\min(p, L) - \max(b, H-p))$, $P' = \min(A, L-p) - b$ and $E' = \min(b, H-p) - (L-p)$,

$$\ell_N(p) = \min\bigl(2n,\ n+R,\ 2R+P'+E'\bigr).$$

These are the exponents of the generic valuation theorems in the formalization; for the primes that interest us they are the same as in OpenAI's paper for its design. For the design D the function $\ell_N(p) = Nf(p/N)$ is piecewise linear in $x = p/N$ ($\lfloor\cdot\rfloor$ rounding aside):

$$f(x) = \begin{cases} 116 & x \le 59/2, \\ 175 - 2x & 59/2 \le x \le 69/2, \\ 106 & 69/2 \le x \le 39, \\ 184 - 2x & 39 \le x \le 81/2, \\ 265 - 4x & 81/2 \le x \le 48, \\ 217 - 3x & 48 \le x \le 69, \\ 148 - 2x & 69 \le x \le 70, \\ 78 - x & 70 \le x \le 78, \end{cases}$$

and $f = 0$ for $x \ge 78$, since for $p > H$ the matrix $\mathcal{A}_N^0 + z\mathcal{L}_N$ is p-integral. Its integral over $[1, 78]$ is $24269/4$, and $f = 116 = 2 \cdot 58$ on $[0, 1]$. Note that $f(x) = 2n_0$ for $x \le 59/2$, which contains all the primes in $(N, 2N]$ and, in particular, the whole range $(16/7, 53/2]$ of the rank deficiencies of Theorem 2.4; there the bound of Proposition 2.5 improves $-\ell_N(p) = -2n$ to $\nu_N(p) - 2n$.

The primes $p \le \sqrt{H}$ are not covered by these bounds. For them we use the cruder bound $6n \log_p H$ for the exponent, which contributes $O(\sqrt{N}\, N \log N) = o(N^2)$ in total.

3.7. **The prime 2.** At $p = 2$ the entries are not integral and the bound comes from the expansion of the determinant into minors, in the manner of [26, §3]. Its exponent is governed by a quadratic form in the numbers m, ℓ of columns of two exceptional types, which is

$$\tfrac{3}{2}m^2 + \tfrac{1}{2}\ell^2 - (H + C - b)\, m - C\,\ell + Cn,$$

and the following identity shows that it is at least $-\mathsf{A}_2$ whenever $m, \ell \ge 0$, $m + \ell \le n$ and $C \ge n$, where

$$\mathsf{A}_2 = \frac{(n + H - b)^2}{8} - \frac{n^2}{2}. \tag{3.6}$$

With $s = n - m - l$ and $m_* = (n + H - b)/4$,

$$\tfrac{3}{2}m^2 + \tfrac{1}{2}\ell^2 - (H+C-b)m - C\ell + Cn + \mathsf{A}_2 = 2(m - m_*)^2 + s\,(C - n + m) + \tfrac{1}{2}s^2 \ge 0.$$

This gives a 2-adic bound $v_2(\Delta_N(i)) \ge -\mathsf{A}_2 - O(N \log N)$, uniform in the integer i. For D, $\mathsf{A}_2 = (58 + 78 - 9)^2/8 - 58^2/2 = 2673/8$ in units of N^2.

3.8. **The common denominator.**

Proof of Proposition 2.6. Let $\kappa_N(p)$ be the largest of the lower bounds for $\nu_N(p)$ of Section 4 that apply at p (and 0 if none does), for primes $p \le H$ with $H < p^2$, and put $e_N(p) = \min(\ell_N(p),\, 2n - \kappa_N(p))$. For the primes with $p^2 \le H$ use the crude bound above, for $p = 2$ the 2-adic bound, and let D_N be the product of $p^{e_N(p)}$ over $p \le H$. By Proposition 2.5 and the bounds of this section, $v_p(\Delta_N(i)) \ge -e_N(p)$ for every integer $i \ge 0$ and every p (for $p > H$ the matrix is p-integral), so $D_N \Delta_N(i) \in \mathbb{Z}$. A prime $q \in (N, 2N]$ has $\kappa_N(q) = 0$, since no window contains it, and $\ell_N(q) = 2n$, so $q^{2n} \mid D_N$ (here $N \ge 79$ guarantees $q > \sqrt{H}$).

For the size of D_N, the sum of $\ell_N(p) \log p$ over the odd primes is, by the prime number theorem in the weighted form

$$\frac{1}{N} \sum_{p} F(p/N) \log p \longrightarrow \int_0^\infty F(x)\, dx \qquad (N \to \infty), \tag{3.7}$$

for bounded, Riemann integrable, compactly supported F, equal to $N^2 \int_0^{78} f(x)\, dx + o(N^2)$ with $\int_0^{78} f = 116 + 24269/4$. The primes in the intervals of Theorem 2.4 lie in $(16N/7, 53N/2]$ and have $\ell_N(p) = 116N$, so their exponent is at most $116N - N\kappa(p/N) + 2$; the sum of $(N\kappa(p/N) - 2)\log p$ over these primes is $N^2 \int \kappa + o(N^2)$ by (3.7). The 2-adic part contributes $\mathsf{A}_2 \log 2\, N^2 + o(N^2)$. □

The polynomial Δ_N has degree at most $n - g = d_0 N$ and takes values in $D_N^{-1}\mathbb{Z}$ at the integers. An integer-valued polynomial of degree m has coefficients in $\frac{1}{m!}\mathbb{Z}$ (Newton's interpolation formula), so

$$Q_N = (d_0 N)!\, D_N\, \Delta_N(X) \in \mathbb{Z}[X], \qquad \deg Q_N \le d_0 N, \tag{3.8}$$

and $\log(d_0 N)! = o(N^2)$. In this way the gain from the rank deficiencies enters through the integer polynomial itself, and not through a separate common factor of its coefficients.

Lemma 3.4. *For all large N and every prime $N < q \le 2N$, the j-th coefficient of Q_N is divisible by q^{2j}.*

Proof. For these primes the matrices $q^2 \mathcal{A}_N^0$ and $\mathcal{L}_N$ are q-integral by Lemma 3.3, and q^{2n} divides D_N. The identity

$$Q_N(X/q^2) = (d_0 N)!\, \frac{D_N}{q^{2n}} \det(q^2 \mathcal{A}_N^0 + X \mathcal{L}_N)$$

shows that $Q_N(X/q^2)$ has q-integral coefficients; its j-th coefficient is $c_{N,j}/q^{2j}$. □

3.9. **Nonvanishing.**

Proposition 3.5. *Let D be a design and suppose that the three $n_0 \times n_0$ matrices B_σ, $\sigma \in \{0, 1, -1\}$, defined below, are invertible. Then for all large primes p and every rational z such that pz has denominator prime to p, $\Delta_p(z) \ne 0$. In particular $\Delta_p(a/b) \ne 0$ whenever $a \in \mathbb{Z}$, $b \ge 1$ and $p^2 \nmid b$.*

The matrices are those of OpenAI's reduction modulo p at $N = p$, with the design in place of 48: let B be the rational $(n_0 + 1) \times n_0$ matrix of the entries of the filtered rational part at $N = 1$ for the rows $0, \dots, n_0$ (the row n_0 is an auxiliary row), and let B_σ have rows $B_r + \sigma B_{r+1}$, $0 \le r < n_0$. For the design D, B has 59 rows and 58 columns.

Proof. Let $p > 4H_0$, with $H_0 = H/N = 78$, so that $H = H_0 p < p^2/4$, and let p be prime to the numerators and denominators of the entries of B and of the three determinants. In the construction at $N = p$ write the row index $r = pr_0 + i$ with $0 \le r_0 < n_0$ and $0 \le i < p$. Since $q = q_0 p$, $C = C_0 p$, $h = h_0 p$, $g = g_0 p$, the Frobenius relation $(1-t)^p \equiv 1 - t^p$ expresses the row polynomials modulo p through those of the design at $N = 1$ and the successor row $r_0 + 1$; this is where the auxiliary row enters, and it is valid for rows on both sides of g because the polynomials are defined through T_{r-g} and U_{r-g-1} for signed indices. The filter satisfies $s^{b_0 p + k}(1 - s)^{q_0 p} = s^{\ell}(s^p)^{b_0 + k_0}(1 - s^p)^{q_0}$ in $\mathbb{F}_p[s]$, so the filtered columns of one residue class use the raw columns $b_0 + k_0, \dots, b_0 + k_0 + q_0$ of the design at $N = 1$ and never change the residue. By [26, Lemma 3.1 and eqs. (31), (35), (36)], which need only $H < p^2/4$, the reduction of $p^2 \mathcal{A}_p^0$ modulo p is then, after ordering the rows by (i, r_0) and the columns by (ℓ, k_0), the matrix $a^{\mathsf{T}} \otimes B_0 + b^{\mathsf{T}} \otimes B_1$, where a, b are the fixed $p \times p$ matrices of OpenAI's identity (51), with $\det a = 2^{-p(p-1)/2}$. The permutation matrix Π in this decomposition has the eigenvalue 0 once and ± 1 each $(p-1)/2$ times, and therefore

$$\det = (\det a)^{n_0} \det B_0 \, \det(B_0 + B_1)^{(p-1)/2} \det(B_0 - B_1)^{(p-1)/2} \neq 0$$

in $\mathbb{F}_p$. Hence $p^{2n} \Delta_p(0)$ is a p-adic unit. For z with pz of denominator prime to p, the difference $p^2 z \mathcal{L}_p = p\,(pz)\,\mathcal{L}_p$ is p-integral with zero reduction (the entries of $\mathcal{L}_p$ are p-integral), so the reduction of $p^2(\mathcal{A}_p^0 + z\mathcal{L}_p)$ is the same invertible matrix and $v_p(\Delta_p(z)) = -2n$. If $p^2 \nmid b$, then $p(a/b)$ has denominator prime to p. □

For the design D the invertibility of B_0, $B_0 + B_1$ and $B_0 - B_1$ is certified by exhibiting their inverses modulo the prime 1009, which exceeds $H_0 = 78$ so that every denominator of B is a unit there; the three determinants are then nonzero rational numbers. The computation modulo 1009 certifies the base matrices; the conclusion for all large primes is a theorem and not an extrapolation from finitely many primes. Nothing in the proof uses the proportions of the design except $2g < n$, and $H < p^2/4$.

4. Rank deficiencies modulo p

In this section we prove Theorems 2.1, 2.2, 2.3 and 2.4. We begin with the four closed formulas, whose proofs show the mechanism in its simplest form, then prove the counting inequality that underlies all of them, describe the further families obtained from it, and finally the certified data for finitely many windows and the resulting function κ for the design D. The principle

is that of the arithmetic method of Hata and Rhin–Viola, in the form that suits a determinant: if the reduction modulo p of $p^2\mathcal{A}_N^0$ has rank at most $n-e$, then by Lemma 3.2 and Proposition 2.5 the determinant $\Delta_N(z)$ gains a factor p^e at every z with denominator prime to p, in particular at every integer. The gain is in the denominator, and by (3.8) it is passed on to the coefficients of Q_N. We therefore need lower bounds for the dimension of the left kernel of $\mathcal{A}_{N,p}^0$, that is, explicit row combinations $c \in \mathbb{F}_p^n$ with $c^{\mathsf{T}}\mathcal{A}_{N,p}^0 = 0$. The tools are the same throughout: the identities for Chebyshev polynomials in characteristic p

$$\begin{aligned} T_p &= X^p, \qquad & U_{p-1} &= (X^2-1)^{(p-1)/2}, \\ T_{kp} &= T_k(X^p), \qquad & U_{kp-1} &= (X^2-1)^{(p-1)/2}\,U_{k-1}(X^p) \end{aligned} \tag{4.1}$$

in $\mathbb{F}_p[X]$ (the last two follow from $T_{kp} = T_k \circ T_p$ and $U_{kp-1} = (U_{k-1} \circ T_p)\,U_{p-1}$), the relation $(1-X)^p = 1 - X^p$, and the digit formula (3.5).

4.1. **Setting.** Fix a design, an integer $N \geq 1$ and an odd prime p with $H < p^2$. For $0 \leq r < n$ put $m_r = r - g$, so that $-g \leq m_r \leq n-1-g$, and $n-1-g \geq g$. In the notation of (3.1),

$$P_r = (1-X)^h \operatorname{rev}_{C-1} T_{m_r}, \qquad D_r = (1-X)^h \operatorname{rev}_{C-1} U_{m_r-1},$$

where $T_{-m} = T_m$ and $U_{-m-1} = -U_{m-1}$, so that $D_g = 0$. For $c \in \mathbb{F}_p^n$ write $S_c = \sum_r c_r \operatorname{rev}_{C-1} T_{m_r}$ and $U_c = \sum_r c_r \operatorname{rev}_{C-1} U_{m_r-1}$, so that $P_c = (1-X)^h S_c$ and $D_c = (1-X)^h U_c$. The lowest degree of $\operatorname{rev}_{C-1} T_m$ is at least $C-1-|m| \geq a := A = b+q+2g$, and that of $\operatorname{rev}_{C-1} U_{m-1}$ is at least $a+1$. Put

$$Q_c = (X-1)^q\, XW P_c, \qquad \widehat{D}_c = (X-1)^q D_c, \qquad W = (1-X^2)^{(p-1)/2},$$

so that, with $\Lambda = g+q+h$, Q_c has its coefficients in $[a+1,\, L+\Lambda+p-1]$ and $\widehat{D}_c$ in $[a+1,\, L+\Lambda-1]$. Recall that a vector c lies in the left kernel of $\mathcal{A}_{N,p}^0$ as soon as the right side of (3.5) vanishes for every filtered column k (for the columns with a carry among the $q+1$ raw columns one splits the sum over the raw columns at the carry; we shall not need this case).

4.2. **Antisymmetric combinations of rows.**

Theorem 4.1. *Let $2g < p$, $H \leq 4p$ and $D_T = \min(g,\, n-1-g-2p,\, C-3p) \geq 1$. Then $\nu_N(p) \geq D_T - (n+q+g+h-3p)_+$.*

The rows used are the differences of rows placed symmetrically about the three centres g, $g+p$ and $g+2p$.

Proof. For $1 \leq d \leq D_T$ and $k = 0, 1, 2$ let $\delta_k(d) = e_{g+kp+d} - e_{g+kp-d} \in \mathbb{F}_p^n$. These rows exist because $d \leq g$ and $g+2p+d \leq n-1$, and the $3D_T$ vectors $\delta_k(d)$ are linearly independent, since their supports are disjoint: the supports lie in the intervals $[kp, kp+2g]$ of row indices, which are disjoint

because $2g < p$. The row $g + kp \pm d$ has $m = kp \pm d$. By $T_{a+b} - T_{a-b} = 2(X^2-1)U_{a-1}U_{b-1}$ and $U_{a+b-1} - U_{a-b-1} = 2T_aU_{b-1}$ together with (4.1),

$$T_{kp+d} - T_{kp-d} = 2(X^2-1)^{(p+1)/2}\, U_{k-1}(X^p)\, U_{d-1},$$
$$U_{kp+d-1} - U_{kp-d-1} = 2\, T_k(X^p)\, U_{d-1}.$$

Put $R_d = 2(1-X)^h X^{C-d} \operatorname{rev}_{d-1} U_{d-1}$, the value of $D_{\delta_0(d)}$. Reversing at degree $C-1$, and using that $\operatorname{rev}_{kp}(T_k(X^p)) = (\operatorname{rev}_k T_k)(X^p)$, one finds, with $T_0 = 1$, $T_1 = X$ and $T_2 = 2X^2 - 1$,

$$D_{\delta_1(d)} = X^{-p} R_d, \qquad D_{\delta_2(d)} = 2X^{-2p} R_d - R_d,$$

$$XWP_{\delta_0(d)} = 0, \qquad XWP_{\delta_1(d)} = X^{-p} R_d - X^p R_d,$$
$$XWP_{\delta_2(d)} = 2X^{-2p} R_d - 2R_d.$$

The polynomial R_d is supported in degrees $[C-d, H-1]$, which lies in $[3p, 4p)$ because $d \le C - 3p$ and $H \le 4p$. So at each residue ℓ it has a single base-p digit $\varrho_\ell(d)$, in position $u = 3$. For $x = (x_0, x_1, x_2) \in \mathbb{F}_p^3$ the response of the combination $\sum_k x_k \delta_k(d)$ at the raw column $Jp + \ell$ is therefore, by Lemma 3.3, $\varrho_\ell(d)\, f_J(x)$ with

$$f_J(x) = x_0\beta_{3,J} + x_1(\alpha_{2,J} - \alpha_{4,J} + \beta_{2,J}) + x_2(2\alpha_{1,J} - 2\alpha_{3,J} + 2\beta_{1,J} - \beta_{3,J}),$$

which depends neither on d nor on ℓ. The raw columns with $J \ge 3$ do not meet the support, since $3p + \ell \ge C - d \ge C - g = L$. Choose $x^* \ne 0$ with $f_1(x^*) = f_2(x^*) = 0$; this is two linear equations in three unknowns. For $c = \sum_d s_d \sum_k x_k^* \delta_k(d)$ the responses at the columns with $J = 1, 2$ vanish, and the columns with $J = 0$, which have $\ell \ge b$, respond by $f_0(x^*) \sum_d s_d \varrho_\ell(d)$, the coefficient of $X^{3p+\ell}$ in $\sum_d s_d R_d$. These are at most $(H - 3p - b)_+ = (n+q+g+h-3p)_+$ linear conditions on $s = (s_d)$. The map $s \mapsto c$ is injective, so the solutions form a space of dimension at least $D_T - (n+q+g+h-3p)_+$ in the left kernel. □

4.3. **A reflection of Chebyshev polynomials.** The first ingredient of the next theorem is a reflection formula modulo p.

Lemma 4.2. *Let p be an odd prime and $\nu = (p+1)/2$. In $\mathbb{F}_p[X]$, for every integer m,*

$$T_{p-m} = X^p T_m - (X^2-1)^\nu\, U_{m-1}.$$

Proof. Both sides satisfy the three-term recurrence in m, so it suffices to check $m = 0$ and $m = 1$. This needs (4.1). For $m = 0$ it is $T_p = X^p$. For $m = 1$ use $T_{p+1} = XT_p + (X^2-1)U_{p-1}$ and $T_{p-1} = 2XT_p - T_{p+1}$. □

Theorem 4.3. *Let $2h \le p+1$, $L \le 3p \le H$, $n + q + g + h \le 3p$ and $2p \le n+q$, and put $n^* = \min(n, 2p+2g+1)$. Then $\nu_N(p) \ge n^* - g - 2p - 1$.*

Proof. We restrict c to the rows $r < n^*$, so that $m_r \le n^* - 1 - g \le 2p + g$. Applying Lemma 4.2 with $m = m_r$ and reflecting at degree $p + C - 1$ gives, for every such row,

$$(1 - X^2)^\nu \operatorname{rev}_{C-1} U_{m-1} - X \operatorname{rev}_{C-1} T_m = -X^{p+1} \operatorname{rev}_{C-1} T_{p-m},$$

and the right side is divisible by $X^{p+C-|p-m|}$, hence by X^L since $|p - m| \le p + g = p + C - L$. Multiplying by $(1 - X)^h$ and summing, for every c supported on these rows

$$X^L \mid (1 - X^2)^\nu D_c - X P_c. \tag{4.2}$$

Let V be the space of such c with $(1 - X^2)^\nu \mid P_c$. Since $(1 - X)^h$ divides every P_r, this is $\nu - h$ conditions at $X = 1$ and ν conditions at $X = -1$, that is, at most $p + 1 - h$ linear conditions. For $c \in V$ write $P_c = (1 - X^2)^\nu Y$ and $\widetilde{Q} = XY$. Since $(1 - X^2)^p = 1 - X^{2p}$,

$$XW P_c = (1 - X^{2p})\, \widetilde{Q}.$$

The polynomial P_c has no coefficients below X^{A^*}, $A^* = C - n^* + g$, and none from X^H on, so $\widetilde{Q}$ is supported in $[A^* + 1,\ H - p - 1]$; and dividing (4.2) by the unit $(1 - X^2)^\nu$ modulo X^L gives $D_c[k] = \widetilde{Q}[k]$ for $k < L$.

Now let $H - 3p \le \ell < L - 2p$. The inequality $L - 2p \le A^* + 1$ is equivalent to $n^* \le 2p + 2g + 1$, and $L - 2p \le p$ since $L \le 3p$. By the supports, $\widetilde{Q}[\ell] = 0$, $\widetilde{Q}[2p + \ell] = 0$ and $\widetilde{Q}[3p + \ell] = 0$, and $D_c[3p + \ell] = 0$ because $3p + \ell \ge H$. So with $\varkappa = \widetilde{Q}[p + \ell]$ the base-p digits of $XW P_c$ at the residue ℓ are $(0, \varkappa, 0, -\varkappa, 0, \dots)$ and those of D_c are $(0, \varkappa, 0, 0, \dots)$. Hence, by Lemma 3.3, the response at every raw column $Jp + \ell$ with such an ℓ is $\varkappa$ times a constant depending on J only: on V these columns, which have $J \in \{0, 1, 2\}$, span a space of dimension at most one for each residue ℓ.

Put $\ell_0 = H - 3p$ and $\ell_1 = L - 2p$; the hypotheses give $\ell_0 \le b \le \ell_1$. For $\ell_0 \le \ell < b$ the raw columns $j \in [b, L)$ with this residue have $J \in \{1, 2\}$, and for $b \le \ell < \ell_1$ they have $J \in \{0, 1, 2\}$. So the saving is $2\ell_1 - \ell_0 - b$, and the number of linear conditions needed to make all responses on the $n + q = L - b$ raw columns vanish on V is $(p + 1 - h) + (n + q) - (2\ell_1 - \ell_0 - b)$. Substituting $\ell_1 = L - 2p$, $\ell_0 = H - 3p$ this is $2p + 1 + g$. The space of solutions in the span of n^* rows has dimension at least $n^* - g - 2p - 1$. □

4.4. **A block of columns: the digits 1 to 5.**

Theorem 4.4. *Let $e = (p + 1)/2 - q - h \ge 0$, $n \le 4p + 2g + 1$, $L \le 6p$, and put $\mu_0 = \max(q, H + q - 6p)$, $\mu_1 = L - 5p$. Assume $b + q \le p + \mu_0$. Then $\nu_N(p) \ge \mu_1 - \mu_0 - e$, that is*

$$\nu_N(p) \ge \min\Big(\frac{p - 1}{2} - g,\ n + b + q + h - \frac{11p + 1}{2}\Big).$$

The first term is the minimum if $6p \le H$ and the second if $6p \ge H$.

Unlike Theorems 4.1 and 4.3, where the conditions are imposed on the rows, the argument concerns a block of columns. We first need an identity.

Lemma 4.5. *In $\mathbb{F}_p[X]$, for every integer k, $2X^p(X^2-1)^{(p-1)/2}T_k-(2X^{2p}-1)U_{k-1}=U_{2p-k-1}$. If $n\le 4p+2g+1$, then for every c, $X^L \mid 2XWS_c-(2-X^{2p})U_c$, and consequently*

$$2\,Q_c[i]=2\,\widehat{D}_c[i]-\widehat{D}_c[i-2p] \qquad (i<L). \tag{4.3}$$

Proof. For all integers m,k one has $U_{m-1}T_k-T_mU_{k-1}=U_{m-k-1}$, since both sides satisfy the same recurrence in k and agree for $k=0,1$. Take $m=2p$ and use (4.1): $T_{2p}=2X^{2p}-1$ and $U_{2p-1}=2T_pU_{p-1}=2X^p(X^2-1)^{(p-1)/2}$. Reflecting at degree $2p+C-1$, the first term becomes $2XW\,\mathrm{rev}_{C-1}\,T_k$, the second $(2-X^{2p})\,\mathrm{rev}_{C-1}\,U_{k-1}$, and the right side $X^{2p+C-1}U_{2p-k-1}(1/X)$. For $k=m_r\ge -g$ this is divisible by X^{C+k}, hence by X^L, if $2p-k-1\ge 0$; and if $2p-k-1<0$ it has degree $k-2p-1$ and is divisible by X^{4p+C-k}, hence by X^L when $k\le 4p+g$, which holds because $k\le n-1-g\le 4p+g$. Summing with weights c_r and multiplying by $(X-1)^q(1-X)^h$ gives (4.3). □

Proof of Theorem 4.4. Let $M=[\mu_0,\mu_1)$ and $w=\mu_1-\mu_0$; we may assume $w>e$. The block consists of the filtered columns $k(J,\mu)=Jp+\mu-(b+q)$ for $1\le J\le 5$ and $\mu\in M$. These indices are non-negative by the hypothesis $b+q\le p+\mu_0$, and smaller than n because $5p+\mu_1-1=L-1$; so there are $5w$ of them. As $\mu<p$, formula (3.5) applies: the response of c at $k(J,\mu)$ is $\sum_u \alpha_{u,J}Q_c[up+\mu]+\beta_{u,J}\widehat{D}_c[up+\mu]$.

Supports. For $\mu\in M$ we have $p+\mu\le p+\mu_1-1=L-4p-1\le a$ by $n\le 4p+2g+1$, so the digits $u=0,1$ of Q_c and $\widehat{D}_c$ vanish. From $\mu_0\ge H+q-6p=L+\Lambda-6p$ we get that the digits $u\ge 6$ of $\widehat{D}_c$ and $u\ge 7$ of Q_c vanish. For $u\le 5$ we have $up+\mu\le L-1$, so (4.3) applies.

From Q to $\widehat{D}$. For $u=2,\dots,5$, (4.3) gives $\widehat{D}_c[up+\mu]=Q_c[up+\mu]+\frac{1}{2}\widehat{D}_c[(u-2)p+\mu]$, and for $u=2,3$ the last index is at most a, so the term vanishes. Hence if $Q_c[up+\mu]=0$ for $u=2,\dots,5$ and all $\mu\in M$, then by induction $\widehat{D}_c[up+\mu]=0$ for $u=2,\dots,5$, and all digits of $\widehat{D}_c$ in M vanish.

A relation from the Frobenius map. We have $Q_c=(-1)^qX(1-X)^{q+h+(p-1)/2}(1+X)^{(p-1)/2}S_c$, and $q+h+(p-1)/2+e=p$. Since $(1-X)^p=1-X^p$,

$$(1-X)^e\,Q_c=(1-X^p)\,Y_c, \qquad Y_c=(-1)^qX(1+X)^{(p-1)/2}S_c,$$

and Y_c is supported in $[a+1,\,C+(p-1)/2]$. Comparing the coefficients at $up+\mu$, subtracting the coefficient at $(u-1)p+\mu$, and summing over $u=2,\dots,6$, the right side telescopes to $Y_c[6p+\mu]-Y_c[p+\mu]$, which vanishes for $\mu\ge\mu_0+e$: the second term because $p+\mu\le a$, the first because $6p+\mu\ge H+q+e=C+(p+1)/2>\deg Y_c$. So, if $Q_c[up+\mu']=0$ for $u=2,\dots,5$ and all $\mu'\in M$, then

$$\sum_{t=0}^{e}(-1)^t\binom{e}{t}Q_c[6p+\mu-t]=0 \qquad (\mu_0+e\le\mu<\mu_1).$$

This is a recurrence with unit leading coefficient: if $Q_c[6p+\mu]=0$ for the first e values $\mu=\mu_0,\dots,\mu_0+e-1$, then it vanishes for all $\mu\in M$.

Counting. Impose on c that the $n-5w$ filtered columns outside the block vanish, that $Q_c[up+\mu]=0$ for $2\le u\le 5$ and $\mu\in M$ ($4w$ conditions), and that $Q_c[6p+\mu]=0$ for the first e values of μ. By the three steps above all the responses of the block columns vanish. These are at most $n-5w+4w+e=n-(w-e)$ linear conditions, so the left kernel has dimension at least $w-e$. Finally, $w-e$ equals $\mu_1-q-e=n+b+q+h-(11p+1)/2$ if $\mu_0=q$ and $(p-1)/2-g$ if $\mu_0=H+q-6p$. □

4.5. **The second window.**

Theorem 4.6. *Let $e=(p+1)/2-q-h\ge 0$ and $n\le 4p+2g+1$, and put*

$$m_0=\max(b+q,\ L+\Lambda-4p),\qquad m_1=\min(p,\ L-3p,\ a+1),$$

where $a=b+q+2g$. Then $\nu_N(p)\ge m_1-m_0-e$.

Proof. Let $M=[m_0,m_1)$ and $w=m_1-m_0>e$. For $\mu\in M$ we have $\mu\le a$, so the digit $u=0$ of Q_c and $\widehat{D}_c$ vanishes; $\mu\ge L+\Lambda-4p$ gives that the digits $u\ge 4$ of $\widehat{D}_c$ and $u\ge 5$ of Q_c vanish; and $3p+\mu<L$, so that (4.3) gives, with $D_i=\widehat{D}_c[ip+\mu]$ and $Q_i=Q_c[ip+\mu]$,

$$Q_1=D_1,\qquad Q_2=D_2,\qquad Q_3=D_3-\tfrac12 D_1.$$

So every response of the filtered matrix at a column with residue $\mu\in M$ is a combination of the four functionals $D_1(\mu),D_2(\mu),D_3(\mu),Q_4(\mu)$ of c. For $\mu\in M$ the columns $k=\mu-(b+q)$, $p+\mu-(b+q)$, $2p+\mu-(b+q)$ and $3p+\mu-(b+q)$ exist, since $b+q\le\mu<p$ and $3p+\mu<L$. So there are at least $4w$ block columns, and their rows lie in the span F of the $4w$ functionals.

We find $w-e$ linear relations among these functionals. As in the proof of Theorem 4.4, $(1-X)^eQ_c=(1-X^p)Y_c$ with Y_c a polynomial. Put $\sigma(\ell)=\sum_{u\ge0}Q_c[up+\ell]$ and sum the coefficients of this identity over the residue class of $\mu\ge e$ modulo p. The right side gives $\sum_u(Y_c[up+\mu]-Y_c[(u-1)p+\mu])=0$, since Y_c is a polynomial, and the left side gives $\sum_{t=0}^{e}(-1)^t\binom{e}{t}\sigma(\mu-t)$. For $\mu\in M$ with $\mu-e\ge m_0$ all $\mu-t$ lie in M, and there $\sigma=Q_1+Q_2+Q_3+Q_4=D_1+D_2+D_3-\frac12D_1+Q_4$. So

$$\sum_{t=0}^{e}(-1)^t\binom{e}{t}\big(\tfrac12D_1+D_2+D_3+Q_4\big)(\mu-t)=0\qquad(m_0+e\le\mu<m_1),$$

and these $w-e$ relations are independent, since $Q_4(\mu)$ occurs with coefficient 1 in the one for μ and only with lower arguments in the others. Hence $\dim F\le 4w-(w-e)$, and the rank of the matrix is at most $(n-4w)+3w+e=n-(w-e)$. □

The proof does not use a determinant or a lower bound for N. Unlike the proof of Theorem 4.4, the telescoping here is over all digits, so no vanishing of Y_c at the ends is needed.

4.6. **The counting inequality.** The four proofs above have one shape. A space of row combinations is chosen, a few linear conditions are imposed, and it is shown that the conditions make every response vanish; the dimension of the space minus the number of conditions is then a lower bound for the nullity. Theorem 2.2 isolates this argument, and allows in addition filters longer than the prime.

Proof of Theorem 2.2. By (3.4), the response of c at the filtered column k is $\sum_{v=0}^{q}(-1)^v\binom{q}{v}R_c(b+k+v)$, the value at T acting as the shift of the polynomial $(1-T)^q$. Modulo p we have $(1-T)^p = 1-T^p$, hence

$$(1-T)^q = (1-T)^r(1-T^p)^a,$$

which is the congruence of Lucas's theorem for the binomial coefficients $\binom{q}{v}$. So the response at k equals $\sum_{u=0}^{r}(-1)^u\binom{r}{u}\,S_c(b+k+u)$, where $S_c(j) = \sum_{v=0}^{a}(-1)^v\binom{a}{v}R_c(j+vp)$; the arguments satisfy $b \le j$ and $j+ap \le b+n-1+q = L-1$. For $c=\iota(x)$ with $x\in\ker\Phi$ we have $S_c(j) = \pm G_x(j)$, and the difference of order r of a polynomial of degree less than r vanishes identically. So $\iota(\ker\Phi)$ lies in the left kernel of $\mathcal{A}^0_{N,p}$, and since ι is injective, $\nu_N(p) \ge \dim\ker\Phi = \dim\mathcal{V} - \operatorname{rank}\Phi$. □

If the conditions only force the responses on a set K of filtered columns to vanish, the same argument, with the remaining $n-|K|$ responses added to the conditions, gives $\nu_N(p) \ge \dim\mathcal{V} - \operatorname{rank}\Phi - (n-|K|)$. The proofs of Theorems 4.1–4.6 are of this form, with K the set of all filtered columns in Theorems 4.1 and 4.3 and a block of columns in Theorems 4.4 and 4.6.

Proof of Theorem 2.3. Since $p\nmid\mathcal{E}$, the reductions of $v_1,\dots,v_d$ are linearly independent, so that V_p is the reduction of V in the coordinates of this basis. The kernel vectors of B_j satisfy an integer identity, so their reductions lie in the kernel of $B_j|_{V_p}$, and they remain independent because their minor is a unit modulo p; hence $\operatorname{rank}(B_j|_{V_p}) \le r_j$. The map $\Phi = (B_j\circ\varphi_{j,k})_{j,k}$ therefore has rank at most $\sum_j w_jr_j$, and its kernel has dimension at least $md - \sum_j w_jr_j$. Theorem 2.2, applied to $\ker\Phi$ with no further conditions, gives the bound. □

In the applications the row combinations are organized in *slot systems*. A slot system consists of data $x_1,\dots,x_D\in\mathbb{F}_p^K$, a linear map $x\mapsto c(x)$ from the data to row combinations, injective on a subspace of admissible data, fixed polynomials $\lambda_1,\dots,\lambda_D\in\mathbb{F}_p[X]$ of degree less than some $e_{\max}$, and two linear maps $\mathcal{T},\mathcal{G}$ from $\mathbb{F}_p^K$ to $\mathbb{F}_p[Y]$, such that for every admissible x the two row polynomials of $c(x)$ multiplied by $(X-1)^h$, respectively by $(X-1)^h(X^2-1)^{(p-1)/2}$, equal $\sum_d\lambda_d(X)\,\mathcal{T}(x_d)(X^p)$, respectively $\sum_d\lambda_d(X)\,\mathcal{G}(x_d)(X^p)$; this is the form in which (4.1) and the relation $(1-X)^p = 1-X^p$ express the row polynomials modulo p. By the digit

formula, the response of $c(x)$ at the raw column $Jp+\ell$ is

$$\sum_{0\le e<e_{\max},\ p\mid H-1-\ell-e} \psi_{J,\,(H-1-\ell-e)/p}\Big(\sum_{d}\lambda_d[e]\,x_d\Big),$$

where $\lambda_d[e]$ is the coefficient of X^e and each $\psi_{J,s}$ is a linear form, a moment form in J and the digit s, determined by $\mathcal{T}$ and $\mathcal{G}$. For the stored subspaces the data are m polynomial coordinates with values in V, and the admissibility of the data is not imposed directly: the jets $\varphi_{j,k}$ are Taylor coefficients at $X=-1,0,1$, coefficients at ∞, and window rows, the map ι from V_p^m to the data is an arbitrary linear map, and only on the joint kernel of the jets is it required to land in the admissible data, to be injective, and to give responses of the form needed in Theorem 2.2.

4.7. **Further families.** The following lower bounds for $\nu_N(p)$ are proved, each for every admissible design, by explicit choices of the space, the conditions and the columns in Theorem 2.2 or its variant above. Each is a piecewise linear function of p and the parameters, positive on windows described by linear inequalities. We describe them in words; the statements are those of the formalization, cited in Section 8.

(i) *Frobenius products.* Row combinations whose polynomial is the product of a fixed factor, a free polynomial P of bounded degree d_P in the Chebyshev variable, and a polynomial $\Lambda(X^p)$ with exponents in a window of K consecutive values; the conditions require Λ to annihilate the functionals seen by the columns, and the bound is $(d_P+1)(K-\#\text{seen})$. For D this gives $53N-2p-1$ on $(26N, 53N/2)$, where $2p+2\le 53N$.

(ii) *A truncated form of Theorem 4.4.* Without the hypothesis $n\le 4p+2g+1$, the bound $\mu_1-\mu_0-e_+-(n-4p-2g-1)_+$ holds whenever $L\le 6p$ and $b+q\le p+\mu_0$; for D it gives $5p-59N+1$ on $(59N/5, 12N]$.

(iii) *A raw tail of the antisymmetric family.* With the four centres $g+kp$ $(0\le k\le 3)$ and $D'_T=n-1-g-3p$, the count is $p-1-(2g+q+h)$ under $4p\le L$, $b\le p$, $D'_T\le g$, $D'_T\le (p-1)/2$, $3p+1+g\le n$ and $4p+b\le H$; for D this is $p-16N-1$ on $(16N, 118N/7]$.

(iv) *Two families of slot polynomials and their sum.* The first fixes a digit $s\le H/p$ and a range of blocks of rows, with slot polynomials divisible by a power of $X-1$; the second has slot polynomials $(X^p-1)P$ with P supported in a window of degrees. Their kernels are independent when both apply, under a separation condition, at the same window s or at the windows s and $s-1$, and the sum of the two bounds is a lower bound. This is the one place where two bounds are added.

(v) *Families with a constant profile on a cell.* Several families (the theorems called MT, M, Z, C and C′ in the formalization) use slot polynomials whose coordinates satisfy finitely many rational linear equations, the *cell* equations; a solution modulo p exists for every p prime to a fixed integer, the modulus of the cell, and the bound is the dimension of the slot space minus the number of conditions. In the families C and C′ the responses depend

only on the residue of the column modulo p, so that the filter costs $(p-q)_+$ conditions; in C′, where $h \geq p$, one writes $(X-1)^h = (X^p-1)(X-1)^{h-p}$, and the factor $(X-1)^{h-p}$ costs $h-p$ more. The family C gives bounds on $(58N/17, 59N/17]$ and $(29N/8, 11N/3]$, and C′ on $(16N/7, 5N/2]$.

(vi) *Block decompositions for $h \geq p$ and $h < p$.* Writing a row index as $kp+d$ and $y = X^p$, the identities

$$(X-1)^h W T_{kp+d} = \big((X-1)T_k\big)(y)\, Q_1 + \big((X-1)(X^2-1)U_{k-1}\big)(y)\, Q_2$$

and its analogue for U_{kp+d-1}, with $Q_1 = (X-1)^{h-p} W T_d$ and $Q_2 = (X-1)^{h-p} U_{d-1}$, hold in $\mathbb{F}_p[X]$ for $h \geq p$; at a residue with a single live digit the responses of all rows of a block are proportional, and the bound counts the blocks. For $h < p$ a difference of order $p-h$ of the coefficients plays the same role. For D these give bounds on $(59N/21, 3N]$, where $h \geq p$, and on $(3N, 16N/5]$.

4.8. **Stored data for finitely many windows.** On the remaining windows below $16.2N$ the bound is that of Theorem 2.3 with a stored subspace. For each of 34 windows we store integer vectors $v_1, \dots, v_d$ spanning V, a set of d pivot columns with the adjugate of the corresponding minor, and for each map B_j its integer matrix, the bound r_j, $d-r_j$ kernel vectors in the coordinates of V, and again a pivot set with an adjugate. A single Boolean function of these data, evaluated by the Lean kernel, checks the shapes, the identities $\mathrm{adj} \cdot v_{\mathrm{piv}} = \delta \cdot I$ and the products $B_j\big(\sum_a k_a v_a\big) = 0$ in integers; its soundness theorem gives the hypotheses of Theorem 2.3 on V and the B_j, with $\mathcal{E}$ the product of the minors. No prime enters this check.

The hypotheses on the jets and on ι depend on N and p. A window is described by finitely many inequalities that are affine in N and $(p-1)/2$, and so are m and the w_j on each of finitely many pieces of the window. A second verified procedure checks on each piece that the numbers of jets and their orders at -1, 0, 1 and ∞ are nonnegative and fit the window, that $p > 59$, that the map ι is injective on the joint kernel and lands in the admissible data of a slot system, and that the response is given on the joint kernel by a polynomial in the column of degree less than q mod p (so that Theorem 2.2 applies), each required inequality in N and $(p-1)/2$ being derived from the window inequalities by sums, nonnegative integer multiples, and division of the variable part by a common divisor with rounding of the constant, that is, by Chvátal–Gomory cuts. Its soundness theorem gives the hypotheses of Theorem 2.3 at every odd prime $p > 59$ of the window with $p \nmid \mathcal{E}$; for large N all primes of the window are of this kind. For 33 of the 34 windows the hypotheses are established in this way, and for the remaining one by a direct proof of the same statement. The windows lie in $(3N, 16.2N)$, where $p > q$, so $a = 0$ in Theorem 2.2.

4.9. **The function κ for the design D.** For D, $L = 70N$, $H = 78N$, $a = 22N$, $\Lambda = 11N$ and $q + h = 6N$. On $(12N, 53N/2]$ the function κ of Theorem 2.4 is as follows, with $x = p/N$.

interval of x	$\kappa(x)$	bound
$(12, 13]$	$x/2 - 5$	Theorem 2.1(c), $(p-1)/2 - g$
$(13,\ 146/11]$	$73 - 11x/2$	Theorem 2.1(c), second term
$(341/25,\ 69/5]$	$11x/2 - 75$	Section 4.7(v), family Z
$(69/5,\ 14]$	$x/2 - 6$	Section 4.7(v), family Z
$(14,\ 711/50]$	$64 - 9x/2$	Section 4.7(v), family Z
$(30287/2000,\ 78/5]$	$7x/2 - 53$	Section 4.7(v), family Z
$(78/5,\ 16]$	$25 - 3x/2$	Section 4.7(v), family Z
$(16,\ 809/50]$	$73 - 9x/2$	Section 4.7(v), family Z
$(809/50,\ 178/11]$	$73 - 9x/2$	Section 4.8
$(178/11,\ 1619/100]$	$x - 16$	Section 4.8
$(1619/100,\ 118/7]$	$x - 16$	Section 4.7(iii)
$(118/7,\ 69/4]$	$9x/2 - 75$	Theorem 2.1(d), $(9p - 2C)/2$
$(69/4,\ 35/2]$	$x/2 - 6$	Theorem 2.1(d), $(p-1)/2 + h - b$
$(35/2,\ 128/7]$	$64 - 7x/2$	Theorem 2.1(d), $(2(n+q+h) - 7p)/2$
$(64/3,\ 23]$	$3x - 64$	Theorem 2.1(a), $3p - (n+q+h)$
$(23,\ 24]$	5	Theorem 2.1(a) up to $x = 70/3$, then (b)
$(24,\ 26]$	$53 - 2x$	Theorem 2.1(b), $n - g - 2p$
$(26,\ 53/2]$	$53 - 2x$	Section 4.7(i)

The six rows on $(341/25, 809/50]$ come from the family Z of Section 4.7(v), in two forms, on $(341/25, 711/50]$ and on $(30287/2000, 809/50]$. The areas of the rows of the table add up to $21.02610\ldots$, and the 84 intervals below 12 contribute $1.03361\ldots$, so that the total is $22.05972\ldots$; the exact sum is computed in the formalization from the list of intervals.

For Theorem 2.1 one checks the hypotheses on the intervals above. The hypotheses $e \geq 0$ and $n \leq 4p + 2g + 1$ both read $x \geq 12$, and $L \leq 6p$ reads $x \geq 35/3$; so Theorems 4.4 and 4.6 apply for $x \geq 12$. For $12 \leq x \leq 13$ one has $\mu_0 = H + q - 6p = 81N - 6p$, so the first branch of Theorem 4.4 applies, and for $x \geq 13$ one has $\mu_0 = 3N$ and the second branch, which is positive until $x = 146/11$. In Theorem 4.6, $m_1 = p$ for $x \leq 35/2$ and $m_1 = 70N - 3p$ above, and $m_0 = 81N - 4p$ for $x \leq 69/4$ and $12N$ above; the value $m_1 - m_0 - e$ is $9x/2 - 75$, $x/2 - 6$ and $64 - 7x/2$ on the three intervals $(50/3, 69/4]$, $(69/4, 35/2]$, $(35/2, 128/7]$, and it is non-positive outside $(50/3, 128/7)$. Theorem 4.1 applies for $x > 10$ and $x \geq 39/2$, and $D_T = \min(5N, 52N - 1 - 2p, 75N - 3p)$; it gives $3p - (n+q+h) = 3x - 64$ up to $x = 23$ and the cap $g = 5N$ up to $x = 70/3$. From $x = 70/3$ the hypotheses of Theorem 4.3 hold up to $x = 26$ (they read $3p \geq L$, $3p \leq H$, $2p \leq n + q$ and $2h \leq p + 1$), and give $\min(n, 2p + 2g + 1) - g - 2p - 1$, which is $5N$ for $x \leq 24$ and $53N - 2p - 1$ above. In each case the value of the theorem is the value in the table minus at most one unit, from the integer rounding. The same kind of check, with linear inequalities in N, p and the parameters, gives the bounds of Section 4.7 on their intervals, and Section 4.8 gives the others. This proves Theorem 2.4; the loss of 2 in its statement covers the integer rounding of all the bounds.

Remark 4.7. Numerical computations of the exact rank of $\mathcal{A}^0_{N,p}$ for several designs and primes show that the bounds of Theorem 2.1 are attained in the windows where they were tested, and that other families add kernel where the windows overlap. These computations are not part of the proof. An upper bound for the nullity is not proved, and none is needed.

Remark 4.8. The subspaces V of Section 4.8 were found numerically, as optimal constant subspaces for the bound of Theorem 2.3 on each window; they are dense rational subspaces with no evident closed form, which is why they are stored. Only the stored integer identities and the checked window conditions enter the proof.

5. The real place

In this section we prove that Δ_N is small not only at G but on an exponentially small interval around it. The statement is Theorem 2.7: for all large N and $|x-G| \le e^{-\rho N}$,

$$|\Delta_N(x)| \le \exp\bigl(-(n_0^2(U - \tfrac{1}{2}\log 2) - \varepsilon)N^2\bigr), \qquad \rho = n_0 c_2 + w.$$

Here again the design is a parameter. We write $\alpha_A = A/n = (b_0 + q_0 + 2g_0)/n_0$, $\beta_h = 2h_0/n_0$, $\gamma_H = H/n = (n_0 + b_0 + q_0 + g_0 + h_0)/n_0$, $\gamma_d = d_0/n_0 = (n_0 - g_0)/n_0$, and

$$c_{07} = 1 - \frac{b_0 + q_0 + g_0}{n_0}, \qquad c_{26} = 3 - \frac{b_0 + q_0 + 2g_0}{n_0}.$$

For D these are $\alpha_A = 11/29$, $\beta_h = 3/29$, $\gamma_H = 39/29$, $\gamma_d = 53/58$, $c_{07} = 41/58$ and $c_{26} = 76/29$. The proof has three parts. We first reduce the perturbation $\Delta_N(x) - \Delta_N(G)$ to determinants in which some columns of $\mathcal{A}_N$ are replaced by values of the row functions at fixed points (§5.1–5.2). We then bound such determinants by OpenAI's energy method, extended to configurations with fewer free points (§5.3). Finally we supply the trial sequences that the energy method needs and verify the resulting inequalities (§5.4–5.5).

5.1. **A quadrature formula for the coefficient of the constant.** Let $\delta = x - G$. Then $\Delta_N(x) = \det(\mathcal{A}_N + \delta\mathcal{L}_N)$, and the question is how large the columns of $\mathcal{L}_N$ are compared with those of $\mathcal{A}_N$. By (3.3) the entries of $\mathcal{L}_N$ are linear functionals of the coefficients of the row polynomials. Since these polynomials have bounded degree, the functionals can be written exactly as finite combinations of point evaluations: if $\tau_1, \dots, \tau_J$ are distinct points and J is at least the number of degrees of freedom of the rows, Lagrange interpolation gives

$$\mathcal{L}_N(r,k) = \sum_{a=1}^{J} A_r(\tau_a)\,\beta_{a,k}, \tag{5.1}$$

where the weights $\beta_{a,k}$ are the values of the same functionals on the Lagrange basis polynomials. This is a Gauss-type quadrature formula, exact

for polynomials of the relevant degree. The point of the formula is that the columns of $\mathcal{L}_N$ become combinations of point evaluations of the same row functions that define $\mathcal{A}_N$, so that the perturbed determinant can again be treated by the energy method. The cost is the size of the weights.

We take $J = d_0 N = 53N$ nodes. Put $g_* = 401/2500$ and $a_* = 2g_*/(1 + g_*^2) = 2005000/6410801$, and let $K = \lceil J/2 \rceil$. The nodes are the points

$$t_{j,\pm} = \pm\sqrt{a_*^2 + (1 - a_*^2)(1 + s_j)/2}, \qquad T_K(s_j) = 0,$$

that is, the roots of $T_K\big((2t^2 - (1 + a_*^2))/(1 - a_*^2)\big)$; we keep all K positive roots and the first $J - K$ negative ones, so that for odd J exactly one root is dropped. They lie in $(-1, -a_*) \cup (a_*, 1)$. In the coordinate x with $t = 2x/(1 + x^2)$ every node has $|x| > g_*$; this matters below, because the energy fields are larger near $x = 0$. Dropping a root changes the nodal polynomial by a linear factor, and the derivative bound, the circle bound and the degree of the Chebyshev polynomial by constant factors only, so an odd number of nodes costs nothing in the exponent. This is needed here, since $d_0 = 53$ is odd and $J = 53N$ is odd for odd N.

Lemma 5.1. *After removal of the common factor of the rows, the weights in* (5.1) *satisfy, for every* $w > 52.788$ *and all large* N*,*

$$|\beta_{a,k}| \le 108000N\, e^{wN}.$$

Proof. At a root, the derivative identity for Chebyshev polynomials bounds $|P_J'(t_{j,\pm})|$ from below by a constant multiple of $Na_*/(1 - a_*^2)$, up to the dropped factor. The weight is a coefficient functional of the Lagrange polynomial, and every coefficient of a polynomial is bounded by its maximum modulus on a circle centred at the origin. On the circle $|z| = r$ with $r = 1001/1000$, write $s = |\operatorname{Re} z|$. For $\omega = (2z^2 - (1 + a_*^2))/(1 - a_*^2)$ one has $|T_K(\omega)| \le (\Theta + \sqrt{\Theta^2 - 1})^K$, where

$$\Theta(s) = \frac{\sqrt{(r^2 + 1)^2 - 4s^2} + \sqrt{(r^2 + a_*^2)^2 - 4a_*^2 s^2}}{1 - a_*^2}$$

is half the sum of the distances from ω to ± 1. The function Θ decreases in s, while the factor $|1 - z|^{q_0 + h_0} \le (r^2 + 1 + 2s)^{N(q_0 + h_0)/2}$ coming from the column filter and from the common factor of the rows increases. A cover of $[0, r]$ by 42 intervals, with rational upper bounds for the square roots at their endpoints, proves, with $d_0 = 53$ and $q_0 + h_0 = 6$,

$$\frac{d_0}{2}\log(\Theta(s) + \sqrt{\Theta(s)^2 - 1}) + \frac{q_0 + h_0}{2}\log(r^2 + 1 + 2s) < 52.788 \qquad (0 \le s \le r),$$

the supremum being 52.7849 at $s \approx 0.125$. The cover is done on the square of this inequality raised to the power N, so that no logarithm is evaluated and the odd value of $d_0 N$ costs only a factor. Since $|z - t_{j,\pm}| \ge 1/1000$ on the circle, the lemma follows. □

The circle of radius 1.001 instead of 1 separates the circle from the nodes; on the unit circle itself the cancellation in the factor $(1-t)^h$ near $t=1$ is what keeps the bound small.

5.2. **Column expansion and an identity of Andréief type.** Expanding $\det(\mathcal{A}_N + \delta\Phi\beta)$, where $\Phi(r,a) = A_r(\tau_a)$, multilinearly in the columns gives a sum over maps f from the columns to $\{\varnothing, 1, \dots, J\}$ of $\prod_{f(k)\neq\varnothing} \delta\beta_{f(k),k} \det M_f$. Here M_f is the matrix whose k-th column is the k-th column of $\mathcal{A}_N$ if $f(k)=\varnothing$ and the vector $(A_r(\tau_{f(k)}))_r$ otherwise.

Lemma 5.2. *If $|\beta_{a,k}| \le B$ and $|\det M_f| \le M c^{m(f)}$ for every f, where $m(f)$ is the number of columns replaced, then*

$$\bigl|\det(\mathcal{A}_N + \delta\Phi\beta) - \det \mathcal{A}_N\bigr| \le M\bigl((1 + J|\delta|Bc)^n - 1\bigr).$$

Each M_f has $p = n-m$ columns that are integrals and m columns that are point values. Andréief's identity extends to this situation.

Lemma 5.3. *Let μ, ν be finite measures, F_r $(r < p+m)$, G_i $(i<p)$ and K integrable functions, and let B be the matrix with $B(r,i) = \iint F_r(t)K(t,s)G_i(s)\,d\mu(t)\,d\nu(s)$ for $i<p$ and $B(r,p+j) = F_r(\tau'_j)$. Then*

$$(p!)^2 \det B = \iint \det\bigl[F_r((t\|\tau')_c)\bigr]_{r,c} \cdot \det[K(t_i,s_j)]_{i,j<p} \det[G_i(s_j)]_{j,i<p}\, d\mu^p(t)\, d\nu^p(s),$$

where $t\|\tau'$ is the list of the p variables followed by the m fixed points.

Proof. For $m=0$ this is Andréief's identity applied twice [12]. In general expand $\det B$ multilinearly in its first p columns, which introduces one pair of variables (t_i, s_i) for each of them, and antisymmetrize first in the t and then in the s variables; the last m columns are constant throughout. □

With $K(t,s) = 1/(1-ts)$ the middle determinant is the Cauchy determinant, and the third is a generalized Vandermonde determinant $\prod_j s_j^b (1-s_j)^q \det[s_j^{E_i}]$. For the latter we use the bound

$$|\det[s_j^{E_i}]| \le \Bigl(\prod_j (1+s_j)\Bigr)^{n-p} \prod_{i<j} |s_i - s_j|, \tag{5.2}$$

valid for distinct exponents $E_i < n$ and $s_j \in [0,1]$. It follows by multiplying the matrix by a unitriangular matrix built from the polynomial $\prod_j (X - s_j)$ and bounding the remaining $(n-p)\times(n-p)$ determinant by the product of the ℓ^1 norms of its columns. The quotient $\det[s_j^{E_i}] / \prod_{i<j}(s_j - s_i)$ is a Schur polynomial, so the bound is close to optimal.

5.3. **The energy estimate with fixed points.** OpenAI bounds the integrand in Andréief's identity by the exponential of $n^2 E$, where E is a discrete logarithmic energy of the points x_i and s_j in the coordinate $t = 2x/(1+x^2)$. The energy consists of an external field for each point, logarithmic interactions $\log|x_i - x_j|$, $\log(1 - x_i x_j)$ and $\log|s_i - s_j|$ among points of the same kind, and an interaction $-\log(1 - 2x_i s_j + x_i^2)$ between the two kinds. There are two cases, according to which of OpenAI's two bounds for the determinant of the row functions is used [26, Props. 5.2, 5.3]; in the second case the interaction among the x_i has weight 2 instead of 1, and in the first the row field acquires a term with a multiplier λ_1 for the constraint that defines the case.

The energy is bounded from above by duality. Given trial sequences $u = (u_k)$ and $v = (v_k)$, one writes the interaction terms as quadratic forms in the moments $\frac{1}{n}\sum_i T_k(x_i)$ and $\frac{1}{n}\sum_j s_j^k$ and completes the squares against u and v. What remains is the sum of the norms $\|u\|^2 = \sum_k u_k^2/k$, $\|v\|^2$ and, for each point, a function of one variable. With $\mathsf{T}(u,x) = \sum_k u_k T_k(x)/k$ and $\mathsf{S}(v,x) = \sum_k v_k x^k/k$ these are, in the second case,

$$X(x) = \alpha_A \log|x| + \beta_h \log(1-x) - \gamma_H \log(1+x^2) - 4\mathsf{T}(u,x) - \mathsf{S}(v,x),$$
$$Y(s) = \tfrac{b_0}{n_0}\log s + \tfrac{q_0}{n_0}\log(1-s) + 2\mathsf{T}(v,s),$$

and in the first case

$$X_1(x) = \alpha_A \log|x| + \beta_h \log(1-x) - (\gamma_H - \tfrac{1}{2})\log(1+x^2) + \lambda_1\big(\tfrac{2g_0}{n_0} - \tfrac{2x^2}{1+x^2}\big) - 2\mathsf{T}(p,x) - \mathsf{S}(v,x),$$

with the same Y (for the first case the trial sequence for the first variables is called p). If $X \le M_X$ and $Y \le M_Y$ on the whole domains, the energy of any configuration is at most a constant determined by M_X, M_Y and the norms. In potential-theoretic terms the extremal configuration is described by an equilibrium measure for two interacting measures in an external field [32], and the best trial sequences are its moments, as Carreira observed [7]; any trial sequences give a valid upper bound.

When m of the n columns are point values, the configuration has only $n-m$ free points of each kind, while the normalization is still by n. The fixed points contribute through the determinant of the row functions evaluated at the concatenated list $t\|\tau'$. The fixed points are of two kinds: a of them are quadrature nodes, with $|x| > g_*$, and b' of them are points $y \in (0,1)$; the numbers of free points, of the first kind and of the second kind add up to n, and we put $\alpha = (a+b')/n$ and $\tau = b'/n$. The missing free points change the energy bound by an amount linear in m, which is the cost of a fixed column. Every step of OpenAI's argument remains valid uniformly in the number of free points: the regularization of the logarithmic interactions, the diagonal corrections, and the completion of squares only involve sums over the points present. For the fixed points of the first kind the relevant field is

the residual field

$$R(x) = -\gamma_d \log(1+x^2) - 2\mathsf{T}(u,x), \qquad g_* \le |x| \le 1,$$

and for those of the second kind it is

$$N(y) = -\gamma_d \log(1+y^2) + \frac{n_0 - 2g_0 - 1}{n_0} \log y - u_1 y, \qquad 0 < y < 1,$$

where u_1 is the first coefficient of u; the coefficient $(n_0 - 2g_0 - 1)/n_0$ is a lower bound, valid for all $N \ge 1$, of $(n-1-2g)/n$.

Lemma 5.4 (budget, second case)**.** *Let $\ell = \log 2$, and let $\mathcal{N}, \Phi, \mathcal{R}, \mathcal{N}_c$ and c_2 be real numbers, with $m \ge 0$, such that*

$$\mathcal{N} + \Phi + U - c_{07}\ell \le -m, \qquad (c_{26}\ell - \Phi + \mathcal{R} - c_2)^2 \le 8\ell m, \qquad \ell - \mathcal{R} + \mathcal{N}_c \le 0.$$

Then for all $\alpha, \tau \ge 0$

$$(-(c_{07} - \tfrac12) + (c_{26}-1)\alpha - \alpha^2 + \tau - \tau^2)\ell + \mathcal{N} + (1-\alpha)\Phi + \alpha(1-\alpha)\ell \\ + (\alpha - \tau)\mathcal{R} + \tau\mathcal{N}_c \le -U + \tfrac12\ell + c_2\alpha.$$

Proof. Subtracting the right side from the left gives $(\mathcal{N} + \Phi + U - c_{07}\ell) + \alpha(c_{26}\ell - \Phi + \mathcal{R} - c_2) - 2\ell\alpha^2 + \tau(\ell - \mathcal{R} + \mathcal{N}_c) - \ell\tau^2$. The last two terms are non-positive by the third hypothesis, and the first three form a concave quadratic in α with non-positive discriminant, since $A + s\alpha - 2\ell\alpha^2 \le 0$ for all α as soon as $s^2 + 8\ell A \le 0$. □

The lemma is applied with $\mathcal{N}$ a bound for $2\|u\|^2 + \frac12\|v\|^2$, $\Phi = M_X + M_Y$, $\mathcal{R}$ a bound for the supremum of R and $\mathcal{N}_c$ a bound for the supremum of N. The analogue for the first case has no fixed points of the second kind, and its polynomial in α is

$$(-(c_{07} - \tfrac12) + (c_{26}-1)\alpha - \alpha^2)\ell + n_1 + (1-\alpha)F_1 + \alpha(1-\alpha)\ell \\ + \alpha\lambda_1\tfrac{2g_0}{n_0} \le -U + \tfrac12\ell + c_1\alpha,$$

with n_1 a bound for the norms of the first trial and $F_1 = M_{X_1} + M_Y$; it holds for all real α when its discriminant is non-positive and $c_1 \le c_{26}\ell - F_1 + \lambda_1\frac{2g_0}{n_0}$ (so that it also holds for $\alpha < 0$).

Combining the energy bounds with Lemma 5.1, a determinant M_f with m fixed columns contributes at most $(5 \cdot 108000N)^m e^{(n_0 c + w)Nm}$ times the unperturbed bound, where $c = \max(c_1, c_2)$; the factor 5^m comes from expanding the row functions at the fixed points in the second case. Lemma 5.2 then gives, for $|\delta| \le e^{-\rho N}$ with $\rho = n_0 c_2 + w$, a factor $(1 + O(N^2))^n = e^{o(N^2)}$ (here w is any number above 52.788 and $c = c_2$). The unperturbed bound is $\exp(n_0^2 N^2(-U + \frac12\log 2 + \varepsilon))$. This proves Theorem 2.7, given the numerical bounds of the next two subsections.

5.4. **The trial sequences.** For the first case we use a trial sequence with 40 Chebyshev coefficients and a finite number of geometric terms, with the multiplier $\lambda_1 = 1.68493357$. For the second case we use trial sequences of the form

$$u_k = c_k + \operatorname{Re}\sum_j r_j z_j^k \qquad (k \geq 1;\ c_k = 0 \text{ for } k > K), \tag{5.3}$$

with $K = 160$ Chebyshev coefficients, dyadic rationals, and 75 geometric terms, 41 of them with complex z_j and complex r_j.

The geometric terms have a simple meaning. Since $\sum_{k\geq 1} z^k T_k(x)/k = -\frac{1}{2}\log(1-2xz+z^2)$, a term (z, r) contributes $-\frac{1}{2}\operatorname{Re} r\log(1-2xz+z^2)$ to the potential $\mathsf{T}(u, x)$. If $x_0+i\epsilon = \frac{1}{2}(z+z^{-1})$ then $1-2xz+z^2 = 2z(x_0+i\epsilon-x)$, so the term is, up to a constant, the logarithmic potential of a point charge of complex strength r at $x_0 + i\epsilon$ and its conjugate. The extremal measure of the energy problem has a gap in its support around $x = 0$, and its potential has a logarithmic singularity at $x = 0$ and square-root behaviour at the edges of the gap. A Chebyshev expansion converges slowly near such points. A charge at a small distance ϵ from them reproduces the local behaviour with one term. We place the complex charges at the 41 equally spaced points $x_0 = -0.16 + 0.008j$ $(0 \leq j \leq 40)$, all with $\epsilon = 10^{-2}$, and the real terms at points $\pm z$ with z increasing from 0.8 to 0.999997; the numbers z_j are rationals with at most twelve decimals. The norms of the geometric terms have closed forms: two terms (z_j, r_j), (z_l, r_l) contribute $-\frac{1}{2}\operatorname{Re}\bigl(r_j r_l \log(1 - z_j z_l) + r_j \bar{r}_l \log(1 - z_j \bar{z}_l)\bigr)$ to $\|u\|^2$.

5.5. **Verification of the bounds.** We must bound three functions of one variable from above on their whole domains: the field X on $[-1, 1]$, the field Y on $[0, 1]$, and the residual field R on $g_* \leq |x| \leq 1$. With $x = \cos\theta$, each is a constant plus a cosine series with K terms plus finitely many terms $\operatorname{Re} w\log(1 - ae^{im\theta})$.

We cover each domain by cells $[x_c-h, x_c+h]$ whose endpoints are multiples of 2^{-40}, and on each cell bound the function by its Taylor expansion at x_c. The cells are grouped in blocks. The polynomial part has degree $K-1$, and a global bound for its fifth derivative would force very short cells. Instead, for each block the derivatives of the polynomial part are computed exactly at the centre of the block up to order 70, and the remainder beyond order 70 is bounded by a single constant for the block, which enters multiplied by $h^{70}/70!$; the Taylor expansion at the centre of a cell is obtained from that of the block by an exact shift. The logarithmic terms are expanded to order 5, with remainders bounded from the exact minimum of $|1 - ae^{im\theta}|$ on the cell. A few cells at $x = 0$ and $x = \pm 1$ treat the singular terms separately. The covers of X, Y and R have 47, 38 and 23 blocks and 373, 302 and 176 cells, respectively. Each cell is checked by exact rational and integer arithmetic.

The resulting bounds are

$$\sup X \le -0.96450499, \quad \sup Y \le -1.57499400, \quad \sup_{g_* \le |x| \le 1} R \le -0.0631062,$$

so that $M_X + M_Y = -2.53949899$; the norms satisfy $2\|u\|^2 + \frac{1}{2}\|v\|^2 \le 0.7736205195$; and the first coefficient satisfies $u_1 \ge 9/50$, which is used near $x = 0$. For the near points, a one-variable estimate shows that $\sup_{0<y<1} N(y) \le -77/100$ for $u_1 \ge 9/50$, with the weight $47/58$ in $N(y)$; the true supremum is about -0.7758. For the first case, with the weights $(\alpha_A, \beta_h, \gamma_H - \frac{1}{2}, \lambda_1, 2g_0/n_0) = (11/29, 3/29, 49/58, \lambda_1, 5/29)$, the bounds $\sup X_1 \le -1.15854$ and $\sup Y \le -1.32981$ are obtained by isolating the zeros of the numerators of the derivatives, which are polynomials with integer coefficients, and bounding the values between them; the norm bound is $n_1 = 0.67761$.

With $c_{07} = 41/58$, $c_{26} = 76/29$ and $U = 2.255858836$, the hypotheses of Lemma 5.4 hold with

$$m = 2.986 \cdot 10^{-6}, \qquad \ell - \mathcal{R} + \mathcal{N}_c \le -0.0137,$$
$$(c_{26}\ell - \Phi + \mathcal{R} - c_2)^2 \le 1.646 \cdot 10^{-5} \le 8\ell m = 1.655\ldots \cdot 10^{-5},$$

with $c_2 = 4.28886$; for the first case $c_1 = 4.09661 < c_2$ is admissible. This gives the cost c_2 of a fixed column, and $\rho = 58c_2 + w = 301.54258$ for $w = 2.9 \cdot 18.203 = 52.7887$ (> 52.788). The decay rate in Theorem 2.7 is $n_0^2(U - \frac{1}{2}\log 2) = 6422.8355\ldots$ The trial data are fixed once and for all; the file containing them has SHA-256 digest

`ff307ca0287dc5224e268202cb0b09776e933652acd2f2f879492af28d3e381d`.

6. Proof of Theorem 1.1

6.1. **The criterion.** We prove Proposition 2.8. Its proof shows where each of the three properties of Q_N enters: the nonvanishing and the smallness are used once at the end, and the divisibility by prime squares is what makes the choice of the prime free.

Proof of Proposition 2.8. Let a/b be a rational number with b large, put $L = \log b$, $U = \kappa L$ and $V = (\kappa - \omega)L$, and suppose $|\xi - a/b| \le b^{-\rho\kappa}$. By the prime number theorem, $(1-\delta)t \le \theta(t) \le (1+\delta)t$ for all large t, and $V \to \infty$ with b. If every prime $p \in (V, U]$ had $p^2 \mid b$, then $2(\theta(U) - \theta(V)) \le \log b = L$; but

$$2\big(\theta(U) - \theta(V)\big) \ge 2L\big((1-\delta)\kappa - (1+\delta)(\kappa - \omega)\big) = 2L\big(\omega - \delta(2\kappa - \omega)\big) > L,$$

by the fourth hypothesis. So there are primes in $(V, U]$ whose squares do not divide b. Let p be the *largest* one. Every prime $q \in (p, U]$ has $q^2 \mid b$, and $q \le U < 2V < 2p$ because $2\omega < \kappa$. Let B be the product of these q^2; then $B \mid b$ and $\log B = 2(\theta(U) - \theta(p))$.

Every term $c_i a^i b^{dp-i}$ of the integer $b^{dp}Q_p(a/b) = \sum_i c_i a^i b^{dp-i}$ is divisible by q^{2dp} for each of these q: q^{2i} divides c_i by hypothesis, and $q^2 \mid b$ gives

$q^{2(dp-i)} \mid b^{dp-i}$. So B^{dp} divides the integer. Since $Q_p(a/b) \neq 0$ (as $p^2 \nmid b$), $b^{dp}Q_p(a/b)/B^{dp}$ is a nonzero integer, and

$$1 \leq \Big(\frac{b}{B}\Big)^{dp} |Q_p(a/b)|. \tag{6.1}$$

Since $\rho p \leq \rho U = \rho\kappa L$, we have $|a/b - \xi| \leq e^{-\rho p}$, so $|Q_p(a/b)| \leq e^{-\eta p^2}$ and (6.1) gives $\eta p^2 \leq dp\big(L - 2(\theta(U) - \theta(p))\big)$, that is,

$$\eta p \leq d\big(L - 2(\theta(U) - \theta(p))\big) \leq d\big(L - 2U + 2p + 4\delta U\big),$$

using $\theta(U) \geq (1-\delta)U$ and $\theta(p) \leq (1+\delta)p \leq (1+\delta)U$ in the form $\theta(U) - \theta(p) \geq U - p - 2\delta U$. Hence $(\eta - 2d)p \leq dL - 2dU + 4d\delta U$. As $\eta < 2d$ and $p \leq U$, the left side is at least $(\eta - 2d)U$, so $\eta U \leq dL + 4d\delta U$, that is, $\eta\kappa \leq d + 4d\delta\kappa$, which contradicts the last hypothesis. □

The choice of the largest admissible prime is what makes the selection free: the primes of the window that are not admissible have their squares in b, and these squares are recovered from the coefficients through the divisibility hypothesis. A cruder choice, of any prime in a short interval that does not divide b, loses a factor. The free lower end $V = (\kappa - \omega)L$ is what allows $\kappa < 2$: the condition $q \leq U < 2p$ for the recycled primes needs only $U < 2V$.

6.2. **The hypotheses for the design D.** Take the design $D = (58, 9, 3, 5, 3)$ and $d = 53$ (the degree of Q_N is at most $53N$, and the polynomials Q_p of the proposition are the Q_N with $N = p$), and

$$\rho = 301.54258, \quad \eta = 30.041, \quad \omega = \tfrac{3}{4}, \quad \delta = 2.9 \cdot 10^{-7},$$
$$\kappa = \frac{532}{\rho} = 1.764261\ldots$$

We check the hypotheses.

- $\deg Q_N \leq 53N$ by (3.8) and Lemma 3.1.
- The nonvanishing: by Proposition 3.5, $\Delta_p(a/b) \neq 0$ for large p with $p^2 \nmid b$, hence $Q_p(a/b) \neq 0$.
- The divisibility: by Lemma 3.4, q^{2i} divides the i-th coefficient of Q_p for all primes $p < q \leq 2p$.
- The smallness: by Theorem 2.7, Proposition 2.6 and $\log(53N)! = o(N^2)$, for $|x - G| \leq e^{-\rho N}$ and large N,

  $$|Q_N(x)| \leq \exp\big(-(6422.8355\ldots - 6392.7880\ldots - 3\varepsilon)N^2\big) \leq e^{-\eta N^2}$$

  since $\eta = 30.041 < 30.04748\ldots$ is rounded down. In the formal proof this is the inequality $\eta < 6422.8355\ldots - \mathsf{A}$ with $\int\kappa$ replaced by its lower bound 22.05325, which needs $22.05325 \geq \eta - 7.98775$.
- The numerical conditions: $\rho\kappa = 532$; $0 < \eta < 2d = 106$; $\omega = 3/4 \leq 1$ and $2\omega = 1.5 < \kappa = 1.764261\ldots$; $2(\omega - \delta(2\kappa - \omega)) = 1.49999\ldots > 1$; and $d + 4d\delta\kappa = 53.000108\ldots < \eta\kappa = 53.000183\ldots$.

Proposition 2.8 therefore gives $|G - a/b| > b^{-\rho\kappa} = b^{-532}$ for all integers a and all sufficiently large b. This proves Theorem 1.1. The last condition is equivalent to $532 > x = d\rho/(\eta - 4d\delta) = 531.99924\ldots$ With the exact gain in place of the rounded rate, the limit $\delta \to 0$ of the argument is $d\rho/(6422.8355\ldots - \mathsf{A}) = 531.88\ldots$; the integer 532 is the one for which all conditions are verified with rational constants.

7. Remarks on the method

The exponent $d\rho/\eta$ has three ingredients. The degree $d = n - g$ and the perturbation exponent $\rho = n_0 c_2 + w$ come from the construction and from the quadrature: of $\rho = 301.54258$, the cost of fixed columns $n_0 c_2$ is 248.75 and the logarithmic size of the weights is 52.79. The decay rate η is a small difference of two large numbers, the energy at the real place, 6422.8355, and the exponent of the denominators, 6392.7880, and the rank deficiencies of Section 4 contribute 22.06 of the 30.041. Since the exponent is inversely proportional to η, an additional gain of 0.01 in the area of κ lowers it by about 0.18.

All the results of Section 4 are lower bounds for the nullity, and the proofs give the kernel vectors explicitly. The closed formulas, the counting inequality and the families derived from it are proved for every design, and so are the other arguments of the paper; the stored subspaces of Section 4.8 belong to the design D. A different design changes d, ρ and η together, and we make no claim that D is optimal. Except for one pair of families, the function κ does not use any relation between different families, and at a prime where windows overlap the exact nullity may be larger than each bound. An upper bound for the nullity is not needed; more independence between families, or larger stored subspaces, would increase η without changing anything else.

8. Formalization

All results of this paper are formalized in Lean 4 [8] with Mathlib [35], on top of OpenAI's formalization of the irrationality of G [26]. The development is available at

https://github.com/slitsey/catalan-irrationality-measure.

Theorem 1.1 is the theorem

`CatalanMeasure.GenD.Uniform.Bank532.catalan_irrationalityMeasure_532`

in `CatalanMeasure/GenD/Uniform/Bank532/Hosts532Inst.lean`. It uses only Mathlib in its statement, with Catalan's constant written as the series $\sum_j (-1)^j/(2j+1)^2$ and the conclusion $|G - a/b| > b^{-532}$ for all integers a and all sufficiently large b. The modules of the development contain no `sorry` and no `native_decide`, and the command `#print axioms`, which ends the file above, is the check that only the three standard axioms `propext`, `Classical.choice` and `Quot.sound` are used. The repository also contains a configuration `comparator.json` for `lake comparator` [21], which checks that a solution file proves a challenge statement with these axioms only.

The prime number theorem is taken from the PNT+ project [36]. The parts of OpenAI's formalization that are used were converted to Lean's module system and ported to a recent Lean release; the repository's `README` and `PORTING.md` describe the changes.

8.1. **Module layout.** The design-generic part of the development is in the directory `GenD` of `CatalanMeasure`. Its design is that of the paper: a design is data, the generic modules prove every statement for an arbitrary design, and the instance files contain numerals only.

Generic layer:
`PL` (piecewise linear profiles, their areas) and `PrimeGrowth` (the weighted prime number theorem (3.7) for ramp profiles; one exponent per prime); `Budget` (the scalar inequalities of Lemma 5.4 and its analogue for the first case, and the conditions of the criterion), `PayW` and `Recycled` (Proposition 2.8 with the free window end); `Final` (the structure `DesignData` and the theorem `measure_of_design`, which contain no numeral).

Denominators:
`OddProfile` (the valuations $\ell_N(p)$ for the lower and upper pools), `TwoAdic` (the bound (3.6)), `DenomExp`, `DenomGrowth`, `DenomGrowth2`, `Denominator` and `Normalizer` (the common denominator, Q_N, the divisibility of Lemma 3.4); `Uniform/Arithmetic` and `Uniform/LedgerW` (the common denominator `canonicalNormalizer` built from an arbitrary lower bound for the nullity, and the growth of its logarithm).

Closed formulas:
`Interface` (the nullity, the four families and `famKap_le_nullity`), `RankCommon`, `RankK6`, `RankDigits`, `RankFam`, `RankFilt`, `RankSeq`, `RankBlock` (the shared layers: Chebyshev reflection, identity (4.3), the digit formula in filtered form, the block framework), and `RankT`, `RankU`, `RankUCore`, `RankU6`, `RankS` for Theorems 4.1–4.6; `KapHelpers` turns the comparison with a window list into linear arithmetic.

Counting inequality and families:
`Uniform/W/`: `Core` (the counting inequality with a set of columns, `w_core`, `w_core_all`), `Profile` (its profile form), `Lucas/PStep` and `Charts3/PStepPoly` (the step of size p: `gKill_of_pstep_poly` and `slot_nullity_pstep_poly`), `Charts3/Host` (Theorem 2.3 as `host_count` and the lower bound `NullityLB.ofPacketHost`), `Slots` (the slot systems), `Uniform/Prefix` (the responses in digit form and the splitting $(X-1)^q = (X-1)^{q \bmod p}(X^p-1)^{\lfloor q/p \rfloor}$ of the filter, `filter_polynomial_split`), and the families of Section 4.7 in `Frob`, `FrobInst`, `Slots`, `SlotsA`, `SlotsT`, `MTGlue`, `HYPlus`, `ZoneZ`, `ZoneZ2`, `Tier3`, `LStar`, `LStar2`, `TheoremC` and `Thin`.

Stored data:
`Uniform/W/Prof/Packet` (the stored subspaces, `Packet.check` and its soundness `Packet.check_sound`) and `Uniform/W/Charts6/Check` (the window conditions: affine forms and their derivations, `checkHost` and

its soundness `physHost_of_checks`), with `Uniform/W/Charts4` and `Uniform/W/Charts5` (the slot systems of the stored subspaces, and the direct proof for the remaining window); the data of the 34 windows are in `Uniform/Bank532/Pk` and `Uniform/Bank532/Host_*`.

The profile of D:

`Uniform/Bank540`, `Uniform/Bank535` and `Uniform/Bank532`: the interval lists with their certificates (`WinCert`), the growth of the denominator from any list of families (`growth_of_families`), the list `wins532` of the 102 intervals with its area `gain532` and the bound `gain532_ge`, the conditions of the criterion (`Bank532/Gate`) and the final assembly `Final532`.

Nonvanishing:

`Nonvanishing` (Proposition 3.5 from the three base matrices), and the residue-table certificates `CertMod`, `ResBase`, `ResCheck`, `ResInv`, `ResRec`.

Real place:

`Analytic/` (23 modules): the constants and fields of a design (`Const`, `Fields`, `CommonField`, and the numerals for D in `Instances`), the first-case and second-case contracts (`First`, `FirstGlue`, `Second`, `Glue`), the grouped products over the fixed points (`GroupedProduct`, `MarkedSheet`, `MarkedSheetOn`, `MarkedMixedMajorant`, `MarkedSecond`), the prefactor, the perturbation by the quadrature nodes (`Prefactor`, `AtomBase`, `WeightedAtom`, `DeltaSmall`, `DeltaChain`), the quadrature formula (5.1) (`FullInterval`), and the nodes and the weight bounds for any number of nodes (`GapNodes`, `GapNodal`, `GapWeights`, `GapCircleGen`). `Cert/` (6 modules) is the generic cell machinery: the evaluator of logarithms (`WLog`, `WLogSpec`), the residual field with a weight (`ResW`), the shifted tables (`PolyShift`) and the assembly of cells and blocks (`CellLoc`, `Assembly`).

The design D:

`D58/` holds the instance at the real place and for the denominators: the constants of the certificate, `Ramps`, `OddProfile` (the profile f), `Normalizer`, `Circle` (the 42-piece cover), `FirstTrial`, `FirstTrialCert`, `FirstBridge` and `FirstCert/` (the first case), the second case (trial data, the covers of X, Y and R in blocks, the norms, the near-field cap) and `C3/` (the three inverses modulo 1009); the constant $\mathsf{A}_2 = 2673/8$ is evaluated from the generic bound in `Uniform/Bank540/Ledger`. `Uniform/Bank540/Decay` collects the constants ρ and $n_0^2(U - \frac{1}{2}\log 2)$ and the smallness on the neighbourhood of G. The other modules of `D58/` that the final assembly uses (the scalar budgets, the nullity bound `Kap` and the block `Final`) are used in the form in which they stand.

The modules of OpenAI's formalization, the generic valuation theorems of the construction (`CatalanMeasure/Parametric`), the modules of `CatalanMeasure` outside `GenD` on which the argument rests (among them the identity of Andréief type with fixed points, the bound (5.2), the column expansion of Lemma 5.2, integer-valued polynomials and the cell

certificates), and two modules of linear algebra shared with the companion paper (`CatalanCore/Uniform/Content`) are used as they are.

8.2. **Certificates.** The proof combines conventional formal proofs with numerical certificates checked by the Lean kernel: the inverses of the base matrices modulo 1009, the cover of Lemma 5.1, the trial sequences of Section 5.4, the covers of Section 5.5, and the stored subspaces, kernel vectors and window conditions of Section 4.8. All are exact rational or integer computations, decided by the kernel without compiled code. The interval arithmetic used to generate the trial data and the covers [17] is not trusted: Lean rechecks every inequality. Four devices keep the certificates small enough for the kernel.

- *Shared Taylor bounds.* The cells of a cover are grouped in blocks. For each block the derivatives of the polynomial part of the field are evaluated once, exactly, at the centre of the block, together with one bound for the remainder of the expansion; every cell of the block obtains its expansion by an exact shift, instead of repeating the evaluation of a polynomial with 160 coefficients.
- *Accuracy per term from a shared table of nodes.* The logarithms are evaluated by the series $\log \frac{1+x}{1-x} = 2x \sum_i x^{2i}/(2i+1)$ with an explicit geometric tail, and the number of terms is chosen for each logarithmic term separately from a table of nodes shared by all the cells, instead of one accuracy for the whole sum.
- *Residue recurrences.* The tables of the moments $M(i,j)$ and $Z(i,j)$ modulo 1009 are not computed from the rational numbers. They are generated by the one-step recurrences that define these numbers, once proved correct, and the kernel checks the recurrence, the filtered rows against the row polynomials and the products of the matrices with their claimed inverses, row by row, in natural numbers.
- 2-*adic normalisation.* Where a recurrence produces integers divisible by large powers of 2, as the Chebyshev and Clenshaw recurrences do, the tables store the quotients by these powers, with one lemma relating the two; this also avoids integers that the kernel's hashing of literals treats as equal.

Disclosure of AI use

The mathematics, the Lean formalization, the certificate generators and the text of this paper were produced by AI systems at the request of the author, who posed the problems, directed the work and is responsible for this paper: Claude Opus 5.5, Claude Sonnet 5.5 and Claude Haiku 5.5 (Anthropic), running in Claude Code, and GPT-6.1 Sol (OpenAI), running in Codex. The underlying irrationality proof for G and its Lean formalization are OpenAI's [26].

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